Module 2: Descriptive statistics

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Lesson 1: Data types and the toolbox

Understand what the values mean before choosing a summary.

What is statistics?

Collect, organize, present, analyze, and interpret data.
Use evidence to answer a question and support decisions.

Statistical displays

Conceptual illustration supplied for the lesson.

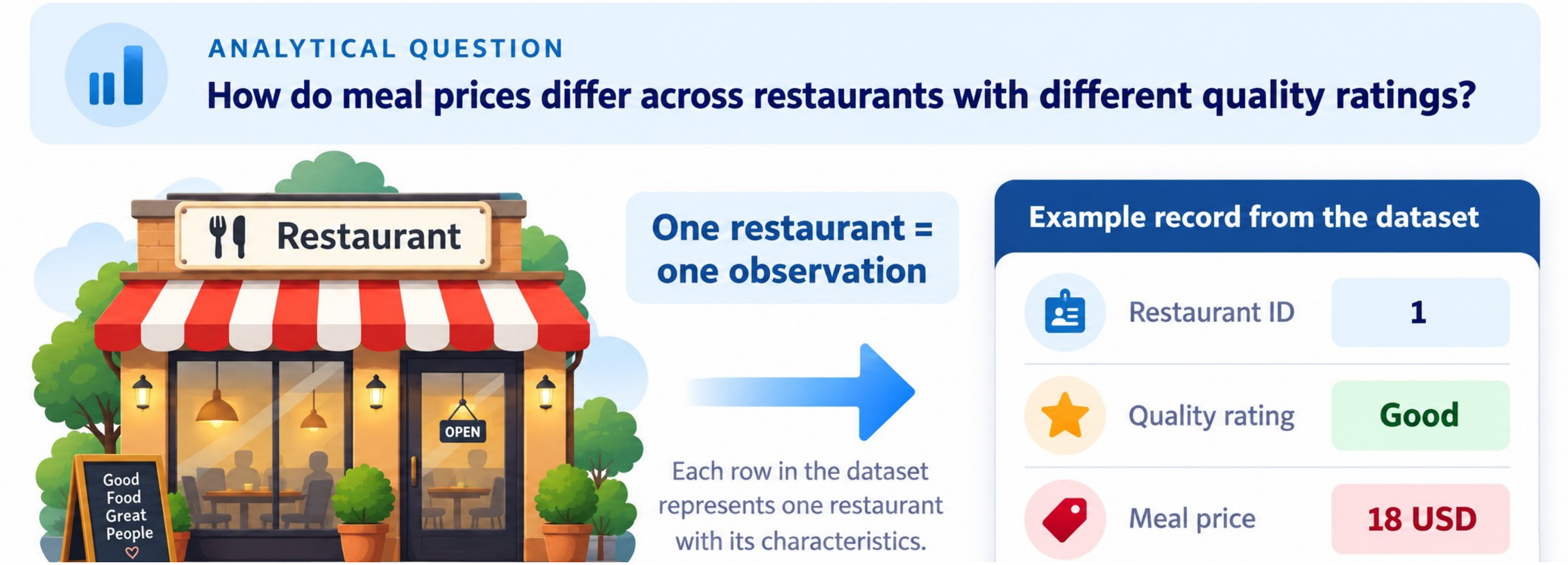


Restaurant case

How do meal prices differ across quality ratings?
First understand each variable, then choose the tools.

كيف تختلف أسعار الوجبات بين فئات تقييم الجودة؟
نفهم أولًا معنى كل متغيّر، ثم نختار أدوات التلخيص.

One restaurant becomes one record



Adapted instructor illustration. The displayed values match the first checked CSV record.

Observation, variable, and unit

Observational unit: the entity being studied.
Variable: a characteristic recorded for that entity.
Observation: one unit’s collection of recorded values.
Here: one row per restaurant; one column per variable.

A number needs a variable definition

101 could be an identifier, a count, or a price.
Its meaning comes from the definition and units.
Numerical appearance alone does not justify arithmetic.

Categorical and numerical data

Categorical values identify a group or ordered category.
Numerical values measure how many or how much.
Measurement level determines which operations are meaningful.

Variable	Examples	Meaning
Payment method	Cash, card, e-wallet	Categorical
Quality rating	Good, Very Good, Excellent	Ordered category
Restaurant ID	101, 102, 103	Identifier
Order count	0, 1, 2, 3	Numerical
Waiting duration	2.4, 5.1, 8.7 minutes	Numerical
Meal price	18, 22, 28 USD	Numerical

Codes remain labels

Cash = 1, Card = 2, E-wallet = 3 are category codes.
A mean code of 2.1 is not an average payment method.
Coding Good, Very Good, Excellent as 1,2,3 does not prove equal gaps.

Discuss: would subtracting telephone numbers answer a measurement question?

Discrete values

Possible values form a finite or countably infinite set.
Counts occur at separate points.
A restaurant can receive 12 or 13 completed orders, not 12.6.

Continuous values

A continuous quantity is modeled as taking any value in an interval.
Waiting time, weight, distance, and temperature are examples.
Between 2 and 3 minutes, 2.1, 2.14, and 2.145 are possible.

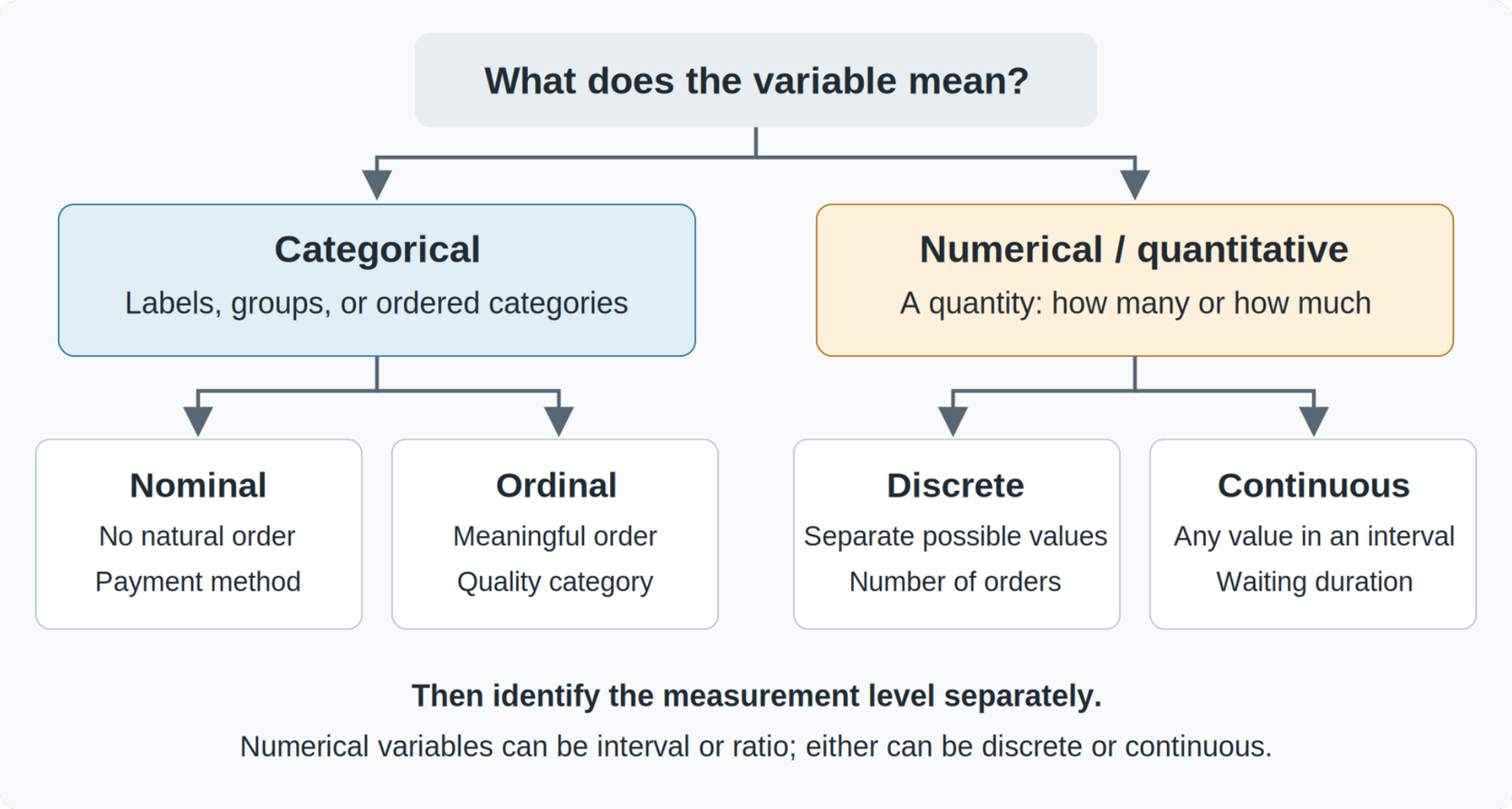
Possible values and recording precision

Decimal places do not decide whether a variable is discrete.
Restricted 0.5, 1.0, 1.5-liter sizes are discrete possibilities.
Audit duration is continuous underneath whole-day recording.
Money recorded to cents has discrete recording increments.

State recording precision and any continuous-model approximation when it matters.

Data-type summary

Classify the variable’s meaning before selecting a summary.



Measurement levels

Measurement level asks which comparisons make sense.
Nominal, ordinal, interval, and ratio add different information.

We explain the four levels before reading their overview diagram.

Nominal: membership

Values label categories without a natural order.
Examples: payment method and restaurant ID.
Use counts, percentages, and category comparisons.

Ordinal: membership and order

Good < Very Good < Excellent is a meaningful ranking.
The gaps between successive categories need not be equal.
Retain the order in frequencies and charts.
A median category can be meaningful; averaging codes adds assumptions.

Interval: meaningful differences

Celsius differences have equal-sized units.
10°C to 20°C is the same change as 20°C to 30°C.
A Celsius mean is meaningful.
20°C is not twice the temperature of 10°C.

Ratio: meaningful differences and ratios

Zero waiting time means no waiting.
20 minutes is twice 10 minutes.
Duration, distance, weight, and order counts are examples.
A meaningful zero still does not permit division by zero.

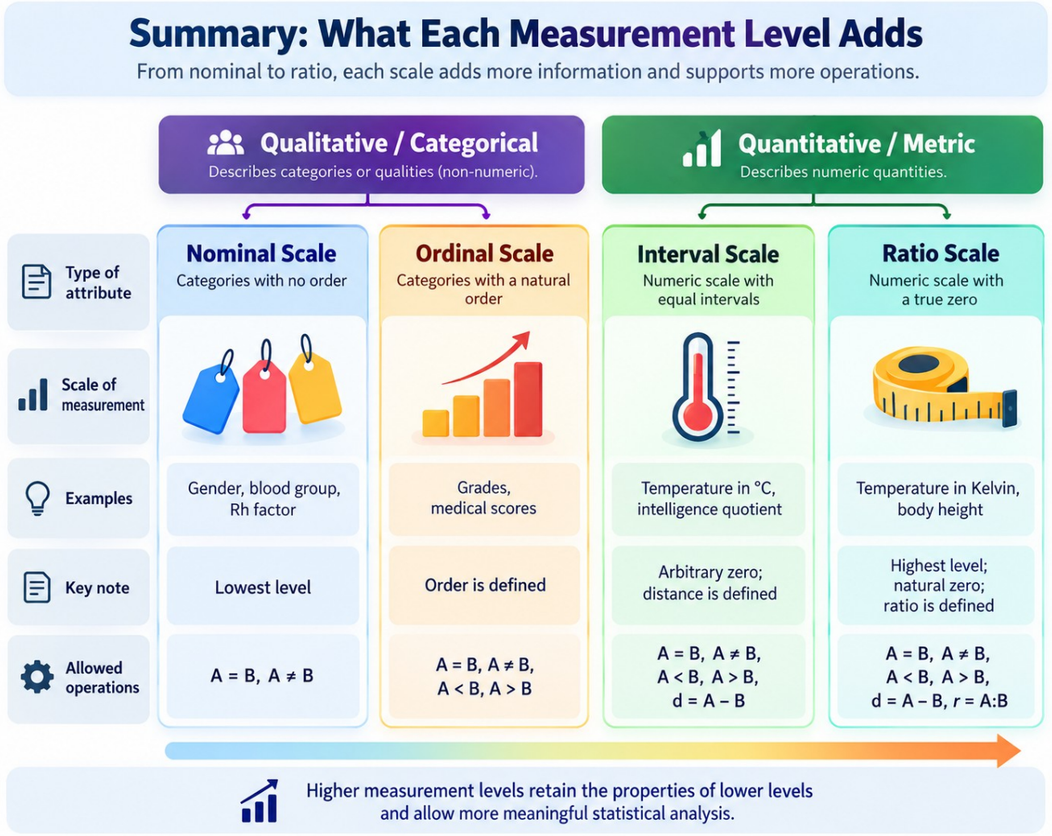
Two classifications, two questions

Discrete/continuous: which values are possible?
Measurement level: which comparisons are meaningful?

Variable	Possible values	Measurement level
Order count	Discrete	Ratio
Waiting duration	Continuous	Ratio
Celsius temperature	Usually continuous	Interval
Recorded calendar year	Discrete	Interval

R storage type is another issue: numeric storage does not prove quantitative meaning.

Your measurement-level overview



Instructor-supplied diagram. Each step adds meaningful comparisons; ratio denominators must be nonzero.

Reading the progression carefully

Nominal distinguishes membership; ordinal adds order.
Interval adds equal differences; ratio adds meaningful ratios.
This progression concerns operations, not the value of an analysis.
Classify “grades” or “medical scores” from their actual definitions.

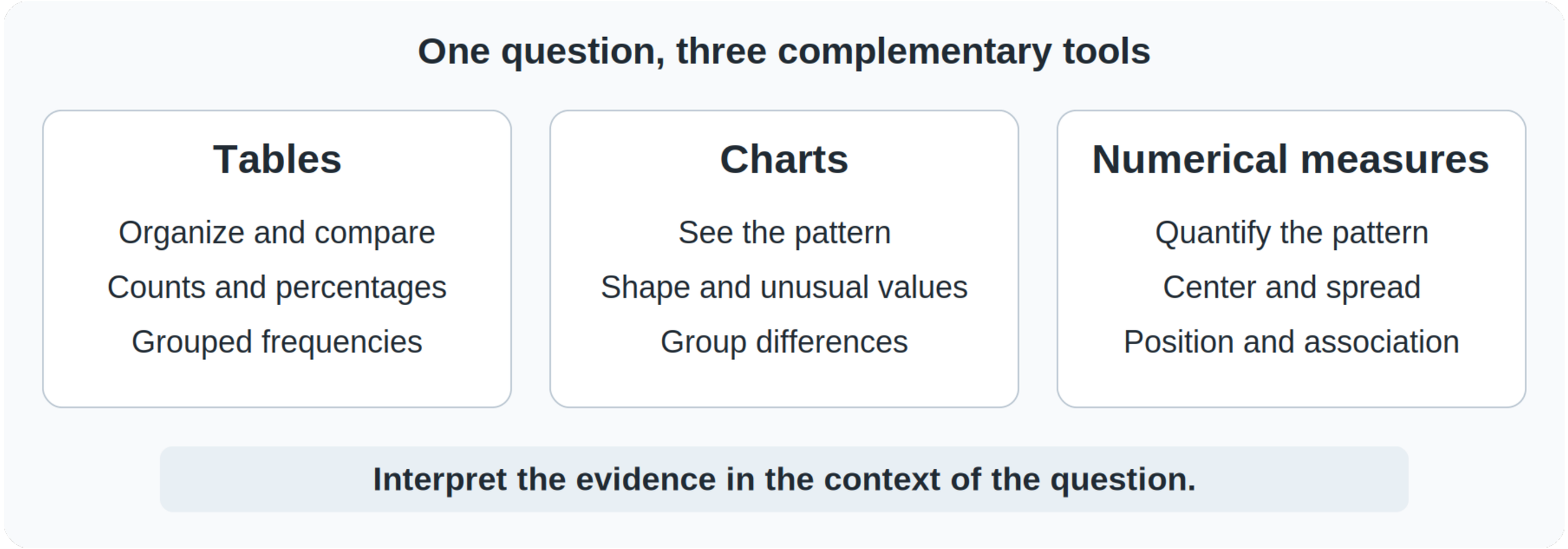
The descriptive statistics toolbox

Describe the observations we have.
Description alone does not establish generalization or causation.



Complementary tools

Tables organize. Charts reveal patterns. Measures quantify features.



Tables organize observations

Frequency tables summarize one variable.
Crosstabs summarize combinations of two categorical variables.
Keep counts, denominators, and units visible.

Frequency and relative frequency

Frequency f: number of observations in a category or class.
Relative frequency = f / n.
Percentage = 100 × f / n.

Quality	Frequency	Relative frequency	Percent
Good	84	0.28	28
Very Good	150	0.50	50
Excellent	66	0.22	22

150 of 300 restaurants (50%) are Very Good; this describes the recorded restaurants.

Audit case

An audit manager asks how long audits take and how much durations vary.
Use 20 recorded audit durations, measured in days.

يسأل مدير التدقيق عن المدد اللازمة للتدقيق ومدى تفاوتها.
نستخدم المدد المسجّلة لعشرين عملية تدقيق، مقيسة بالأيام.

Grouped numerical frequencies

Classes group nearby numerical values.
[15,20) includes 15 and excludes 20.
20 belongs to [20,25). Avoid gaps and overlaps.

Audit duration class (days)	Frequency
[10,15)	4
[15,20)	8
[20,25)	5
[25,30)	2
[30,35)	1

Cumulative counts answer ordered threshold questions; crosstabs count combinations.

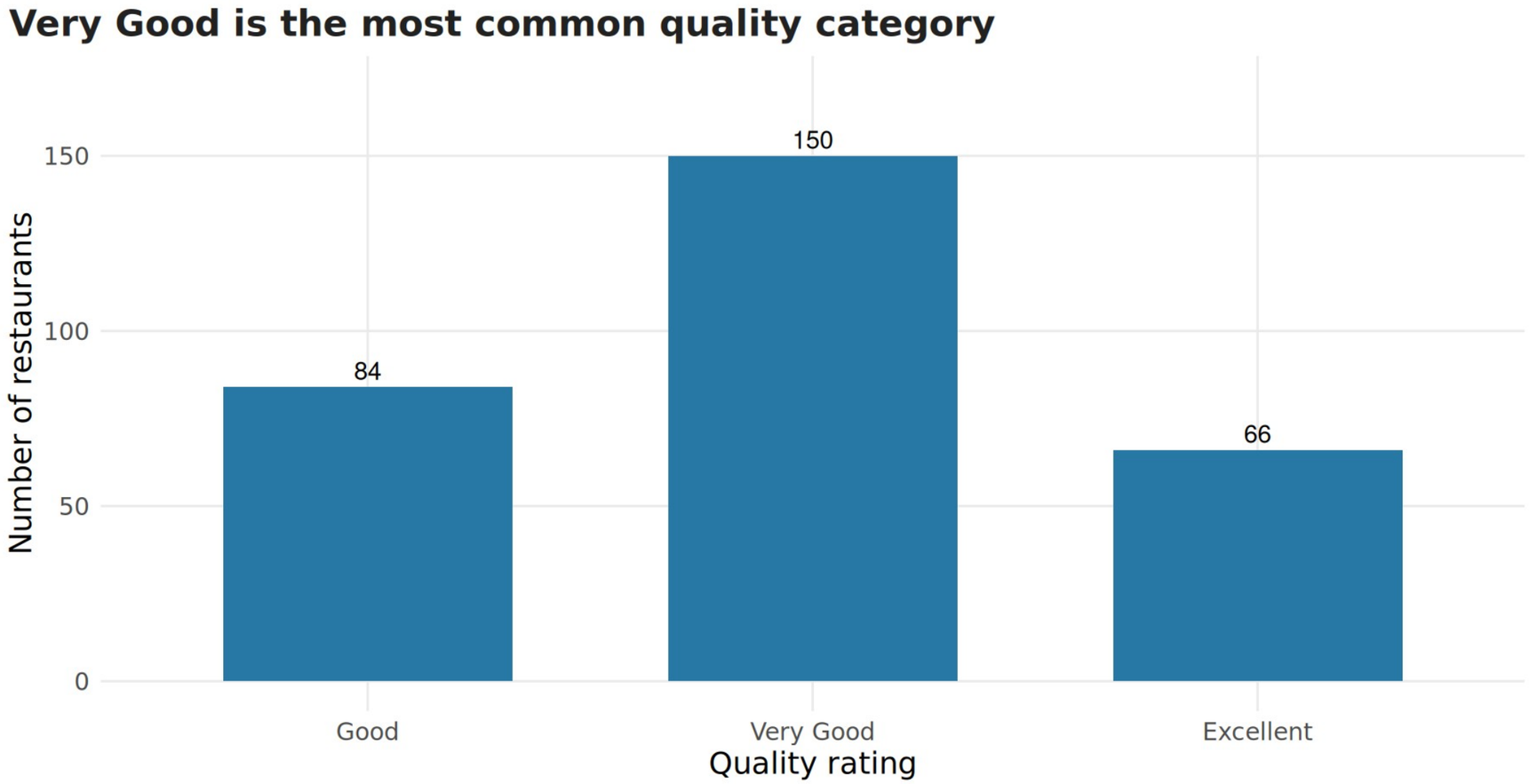
The question chooses the chart

One category variable: bar chart, sometimes a pie chart.
One numerical distribution: histogram or dot plot.
Numerical values across groups: aligned boxplots.
Two numerical variables: scatterplot.
Values over time: a time-series line chart.

Separate category bars; adjacent histogram intervals. Avoid misleading 3D decoration.

An ordered categorical display

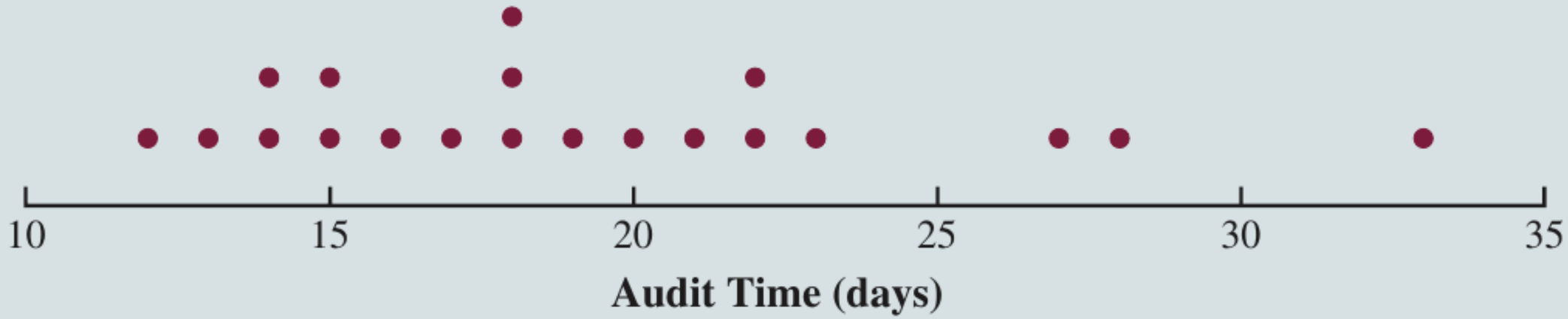
Preserve the natural Good, Very Good, Excellent order.



Every audit remains visible

Each dot is one audit. Stacks show repeated durations.

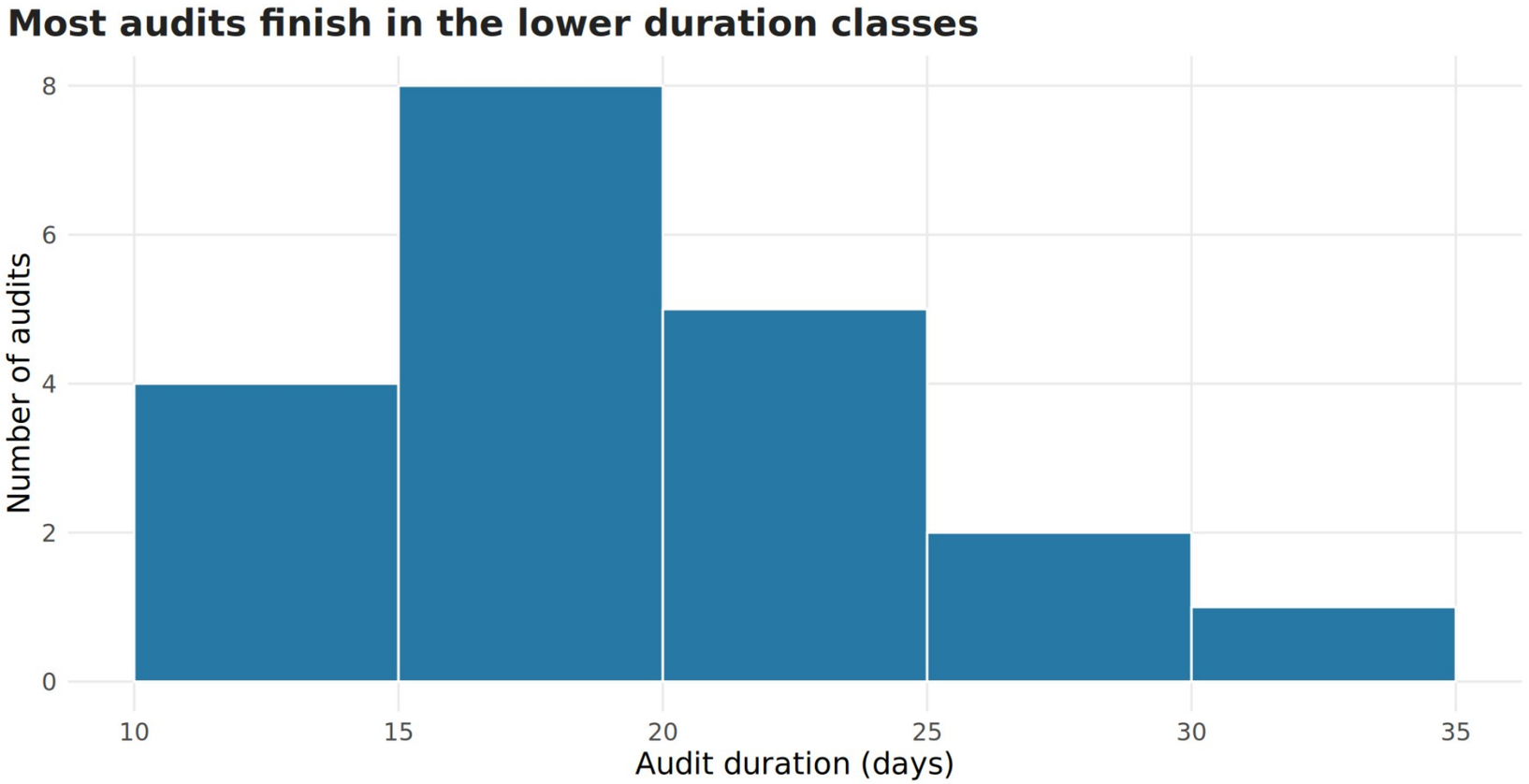
FIGURE 2.11 Dot Plot for the Audit Time Data



Anderson et al., 8th ed., Figure 2.11, p.52. © 2020 Cengage Learning. Attributed excerpt.

The same audits, grouped

The histogram uses the grouped table’s 5-day boundaries.



The 33-day audit remains in the data; investigate its context before any exclusion.

What does a number summarize?

Center: mean, median, mode.
Variability: range, variance, SD, IQR.
Position: percentiles, quartiles, z-scores.
Association: covariance and Pearson correlation.
Shape and unusual values need a plot and context.

Location summaries

Mean: add all values and divide by their count.
Median: middle value or middle-pair average after sorting.
Mode: most frequent value or category, possibly tied.

Arbitrary category codes do not support meaningful means or standard deviations.

Spread and position summaries

Range = maximum − minimum. IQR = Q3 − Q1.
Sample variance divides squared deviations by n − 1.
Sample SD is its square root, in original units.
CV expresses SD relative to a suitable positive ratio-scale mean.
A z-score expresses distance from the mean in SD units.

The Audit numerical overview

These measures describe different features of the same 20 audits.

Measure	Value
Observations	20
Mean (days)	19.25
Median (days)	18
Minimum (days)	12
Maximum (days)	33
Range (days)	21
Q1 (days)	15
Q3 (days)	22
IQR (days)	7
Sample variance (days squared)	29.57
Sample SD (days)	5.44

Quartiles use type = 6. Center 19.25/18 days; SD ≈ 5.44 days; Q1/Q3 = 15/22 days.

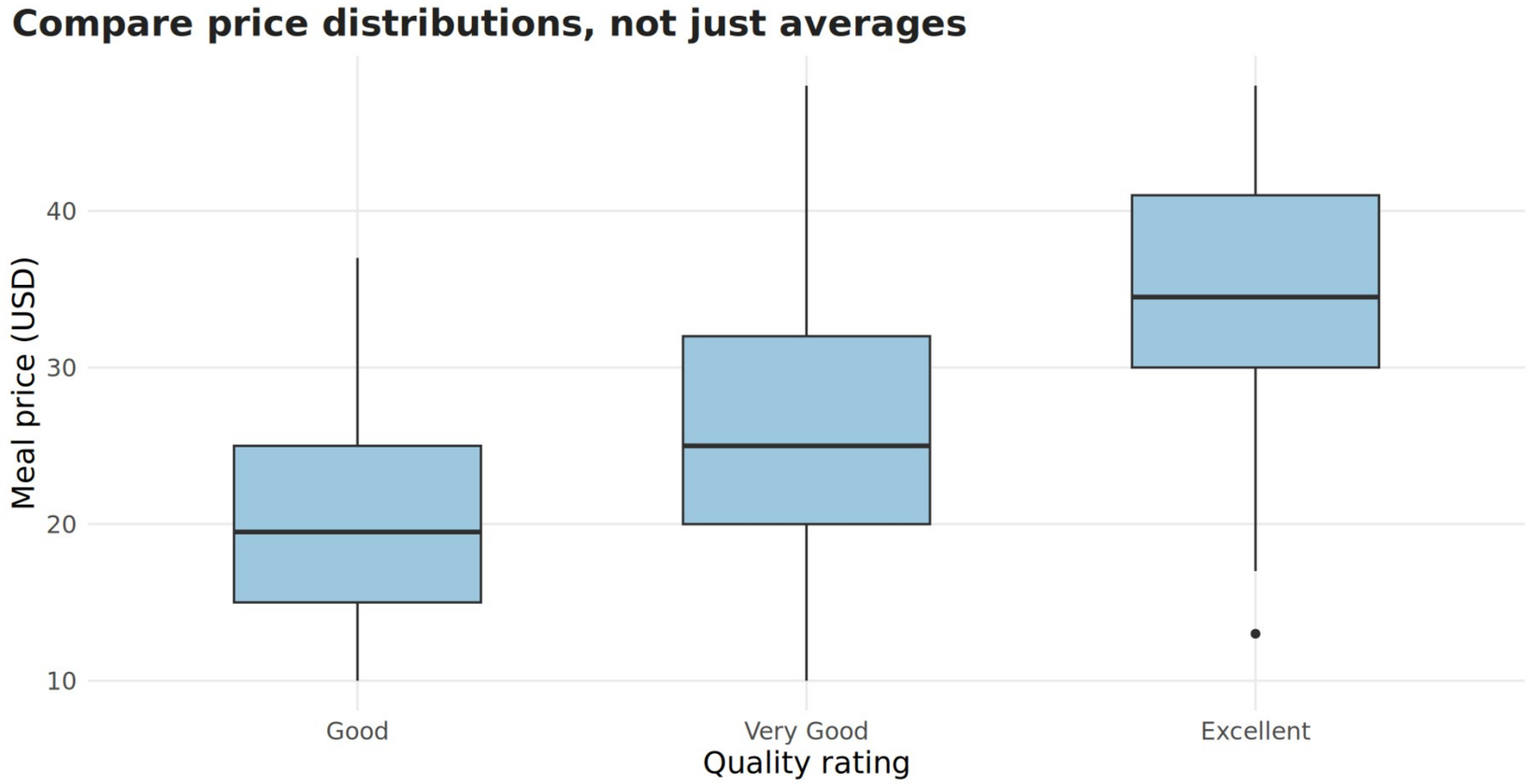
State the percentile convention

The book uses position (n + 1)p / 100 and interpolation.
R’s type = 6 follows this convention for the interior percentiles here.
R defaults to type = 7, so specify the method.

```
# quantile() calculates percentiles; c() supplies the requested probabilities.
# type = 6 follows the book convention; names = FALSE omits percentile labels.
quantile(x, c(.25, .50, .75), type = 6, names = FALSE)
```

Return to the Restaurant question

Compare medians, spreads, overlap, and group sizes.



Overview plot uses ggplot2 boxplot defaults. Lesson 3 aligns plots with explicit type-6 quartiles.

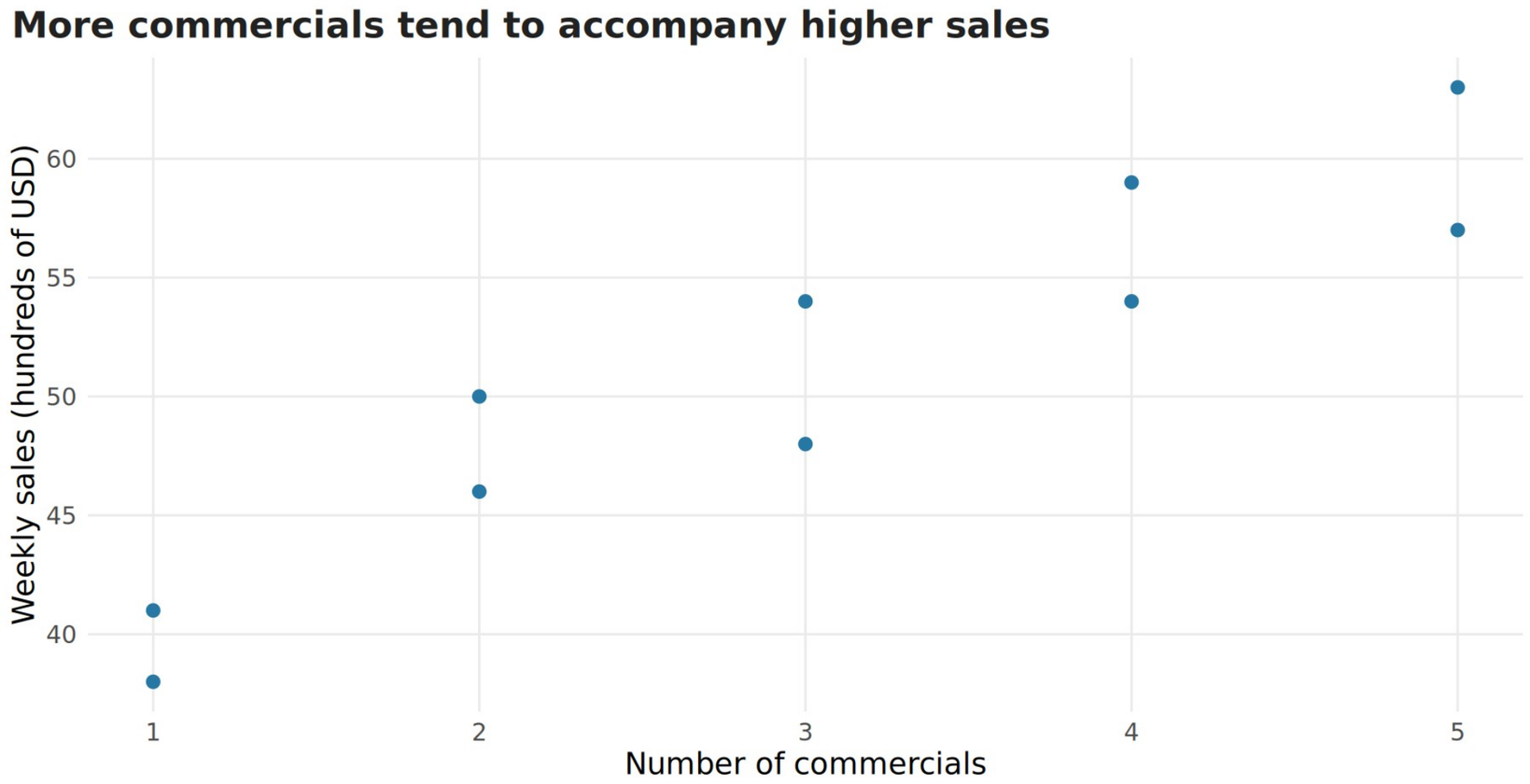
Electronics case

Do weeks with more commercials tend to have higher sales?
Each of ten rows pairs one week’s commercials and sales.
Sales are recorded in hundreds of USD.

هل تميل الأسابيع ذات الإعلانات الأكثر إلى تسجيل مبيعات أعلى؟
يجمع كل صف من الصفوف العشرة عدد الإعلانات ومبيعات الأسبوع نفسه.
سُجّلت المبيعات بمئات الدولارات الأمريكية.

Paired numerical observations

Describe association, then state what the data cannot establish.



A near-zero linear correlation can coexist with a nonlinear relationship. Association does not establish causation.

A short analytical checklist

Define the observational unit, units, and precision.
Classify the variables and measurement levels.
State the question, then choose matching summaries.
Interpret the finding and its limitations.

Pause for the Lesson 1 Quarto exercises.

Data and source scope

Use the prepared CSVs and their data guide.
Restaurant: 300 records; Audit: 20; Electronics: 10.
SoftDrink is developed next.
These historical teaching records do not describe today’s market.

Textbook checks and attribution

Main reference: Essentials of Modern Business Statistics with Microsoft Excel, 8th ed.
Audit: all 20 printed values match. Electronics: all 10 pairs match.
Restaurant: first 10 printed rows and all 12 crosstab cells match.
Remaining individual Restaurant records are not printed for comparison.

Lesson 2: Frequency tables and charts

Turn observations into a distribution, then explain the display.

From observations to interpretation

Count categories and construct numerical classes.
Choose and read the corresponding displays.
Explain shape, center, spread, and unusual features.
Keep denominators and units explicit.

SoftDrink case

Which brands were selected in 50 recorded purchases?
One observation is one purchase, not necessarily one distinct customer.

ما العلامات التجارية المختارة في خمسين عملية شراء مسجّلة؟
تمثّل كل مشاهدة عملية شراء واحدة، وليس بالضرورة عميلًا مختلفًا.

Use brand_book_label. Documented name mapping preserves the original purchase counts.

A valid frequency distribution

Each observation belongs to exactly one category.
Use consistent labels and handle missing values explicitly.
Do not count a record twice. Relevant zero-count categories may remain.

Count, proportion, and percentage

Relative frequency = f / n. Percentage = 100f / n.
0.38 is 38%, not 0.38%.
The totals here are 50, 1, and 100%.

Brand	Frequency	Relative frequency	Percent
Coca-Cola	19	0.38	38
Diet Coke	8	0.16	16
Dr. Pepper	5	0.10	10
Pepsi	13	0.26	26
Sprite	5	0.10	10

Coca-Cola: 19/50 = 38%. Compute Pepsi’s share from 13 purchases before checking Quarto.

Frequency calculations in R

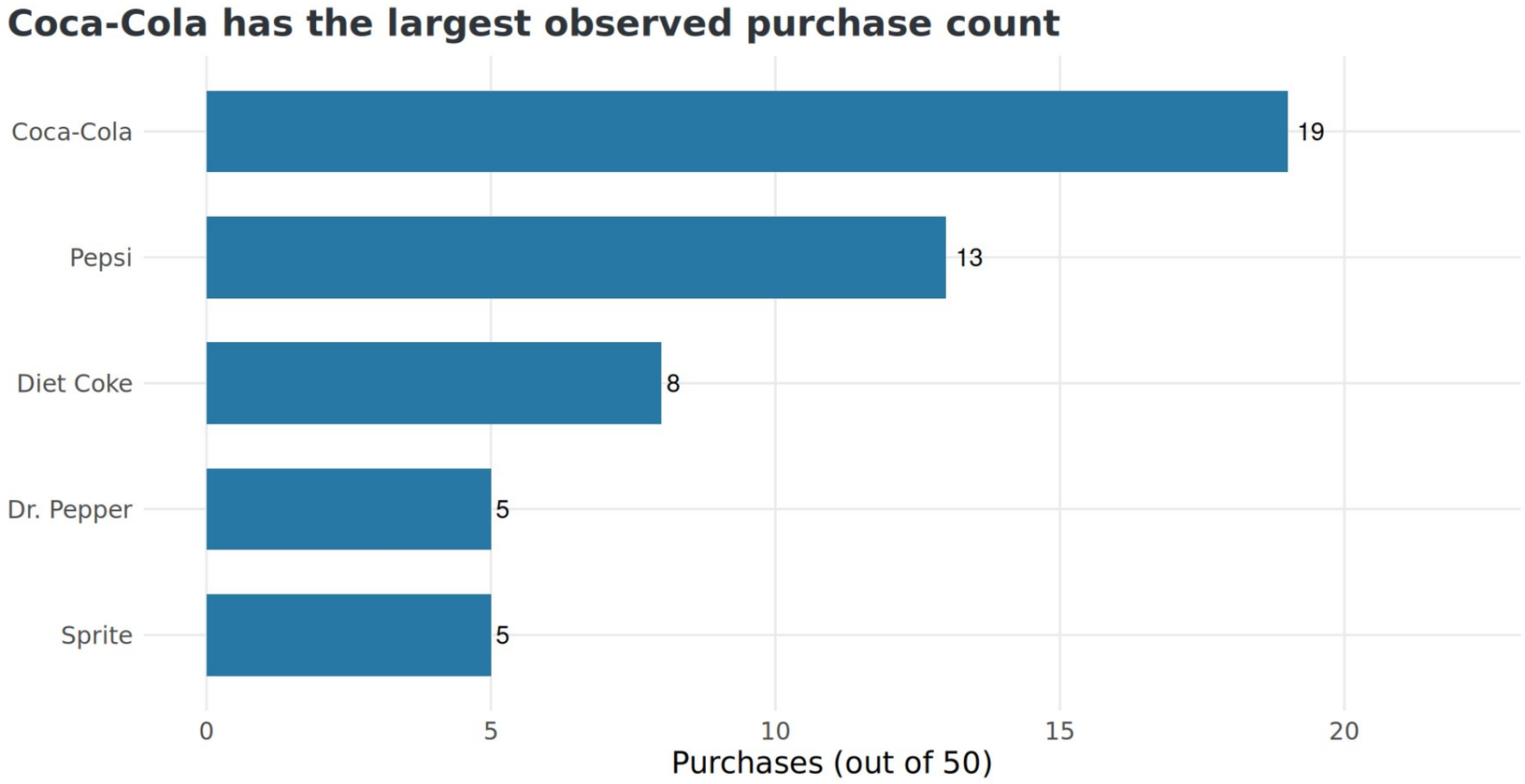
Counts become proportions only after choosing the denominator.

```
# table() counts labels; as.data.frame() makes columns; names()/c() label them.
# nrow(soft) is the 50-purchase denominator; multiplying proportions by 100 gives percentages.
brand_table <- as.data.frame(table(soft$brand))
names(brand_table) <- c("Brand", "Frequency")
brand_table$Relative_frequency <- brand_table$Frequency / nrow(soft)
brand_table$Percent <- 100 * brand_table$Relative_frequency
```

Calculate from unrounded values; round the display only.

Bar charts compare categories

Descending frequency aids comparison of these nominal brands.



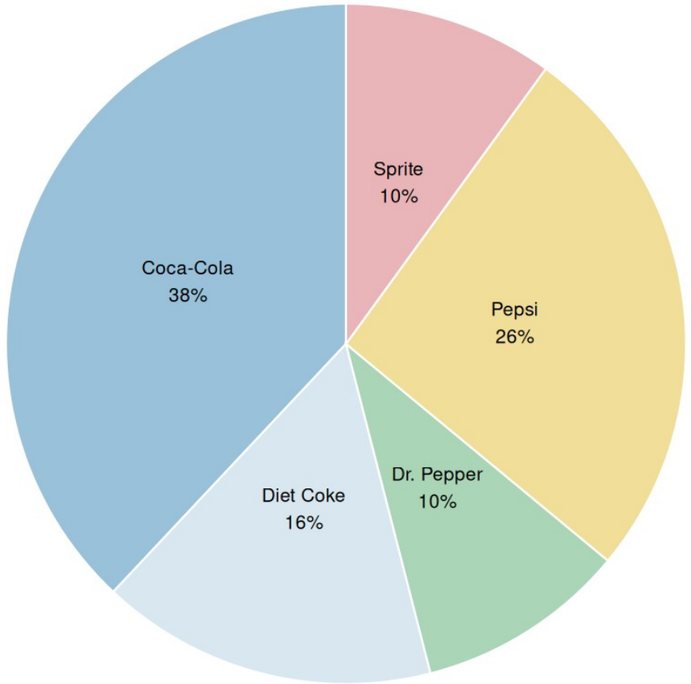
Start the quantity axis at zero. Retain meaningful order for ordinal categories.

Ordering and interpretation

Alphabetical order helps lookup; descending counts help comparison.
Counts and percentages from one sample are rescalings of the same finding.
Observed purchase shares do not establish market share or profit.

Pie charts describe one whole

Categories must be nonoverlapping parts of a named total.



Coca-Cola’s 38% is 136.8°. Discuss the tied shares and long labels; bars often make them easier to compare.

Percentages need a denominator

Name the whole: 50 recorded purchases.
Different sample sizes and sampling coverage matter in comparisons.
Nominal categories have no natural cumulative order.
The top three shares total 80%; “top three” is an analytical choice.

Audit-duration case

Describe 20 year-end audit completion times, recorded in days.
Which ranges are common, and how many finish before a threshold?
One observation is one audit.

نصف مدد إنجاز عشرين عملية تدقيق لنهاية السنة، والمسجّلة بالأيام.
ما فئات المدة الأكثر شيوعًا، وكم عملية انتهت قبل حد محدد؟
تمثّل كل مشاهدة عملية تدقيق واحدة.

These observations do not promise a future client’s completion date.

Numerical classes

The observed minimum and maximum are 12 and 33 days.
Five classes suggest width $(33 - 12)/5 = 4.2$ days.
Use convenient 5-day classes starting at 10.
Grouping reveals patterns but loses individual detail.

Boundary rules

Use [10,15), [15,20), [20,25), [25,30), [30,35).
Include lower endpoints and exclude upper endpoints.
15 belongs to the second class; 20 belongs to the third.
Every value must enter exactly one class, including the maximum.

Audit grouped frequencies

The modal class is 15–<20 days: 8 audits, or 40%.
A modal class does not identify an exact modal duration.

Days	Frequency	Relative frequency	Percent
[10,15)	4	0.20	20
[15,20)	8	0.40	40
[20,25)	5	0.25	25
[25,30)	2	0.10	10
[30,35)	1	0.05	5

For whole-day records, these reproduce the book’s 10–14, 15–19, and subsequent classes.

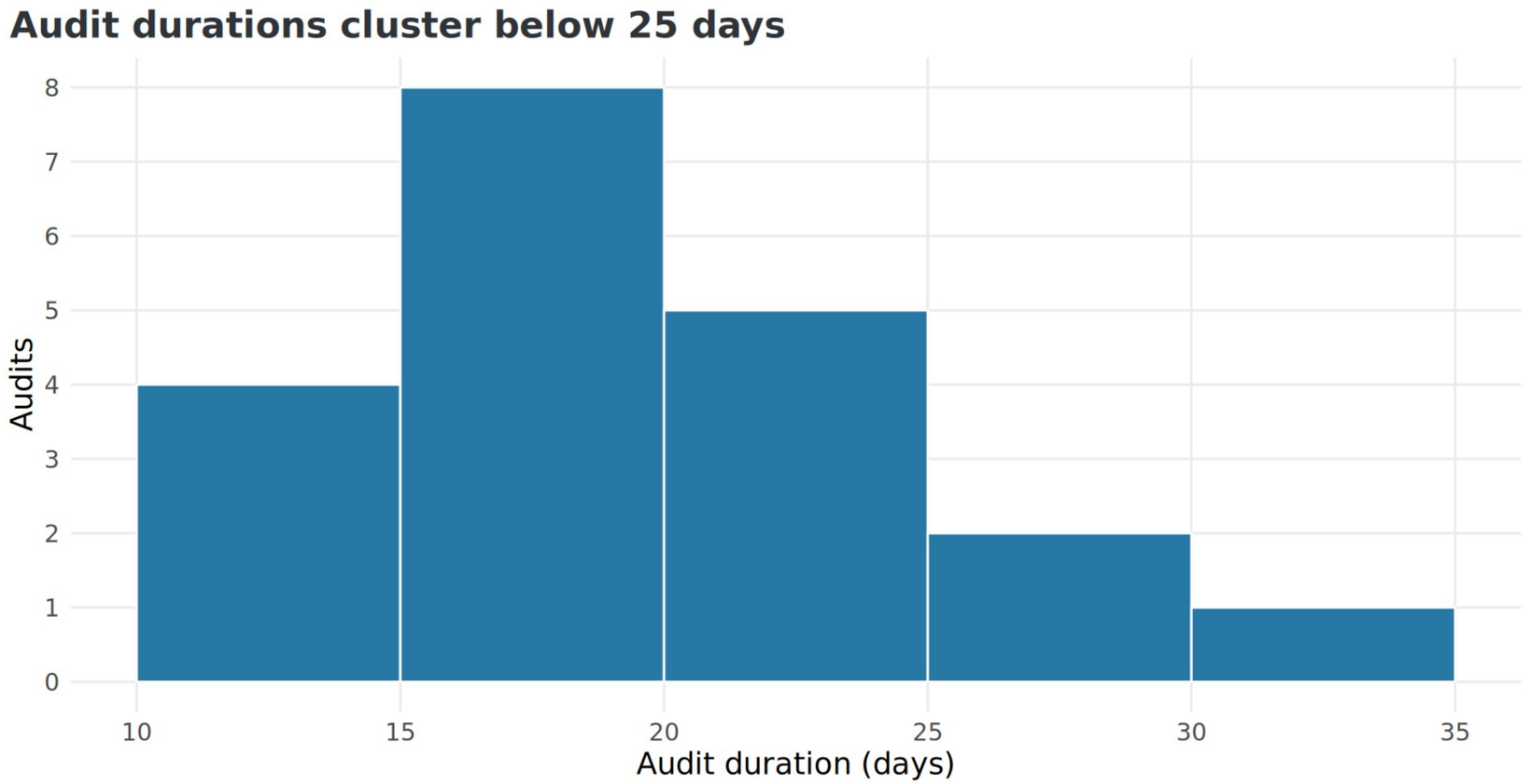
Class construction in R

Choose boundaries deliberately before counting.

```
# seq() creates the 5-day boundaries; cut() assigns each duration to a class.
# right = FALSE gives [lower,upper) intervals; table() counts the resulting classes.
boundaries <- seq(10, 35, 5)
audit$class <- cut(audit$audit_days,
  breaks = boundaries, right = FALSE)
table(audit$class)
```

Audit histogram

Equal-width rectangles count observations in neighboring numerical intervals.



Most audits lie below 25 days. A longer observation warrants context, not automatic deletion.

What does a distribution describe?

Where observations concentrate, spread out, or leave gaps.
The x-axis shows measured values. The y-axis shows its labeled quantity.
A histogram is not a time trend unless time itself is the measured variable.

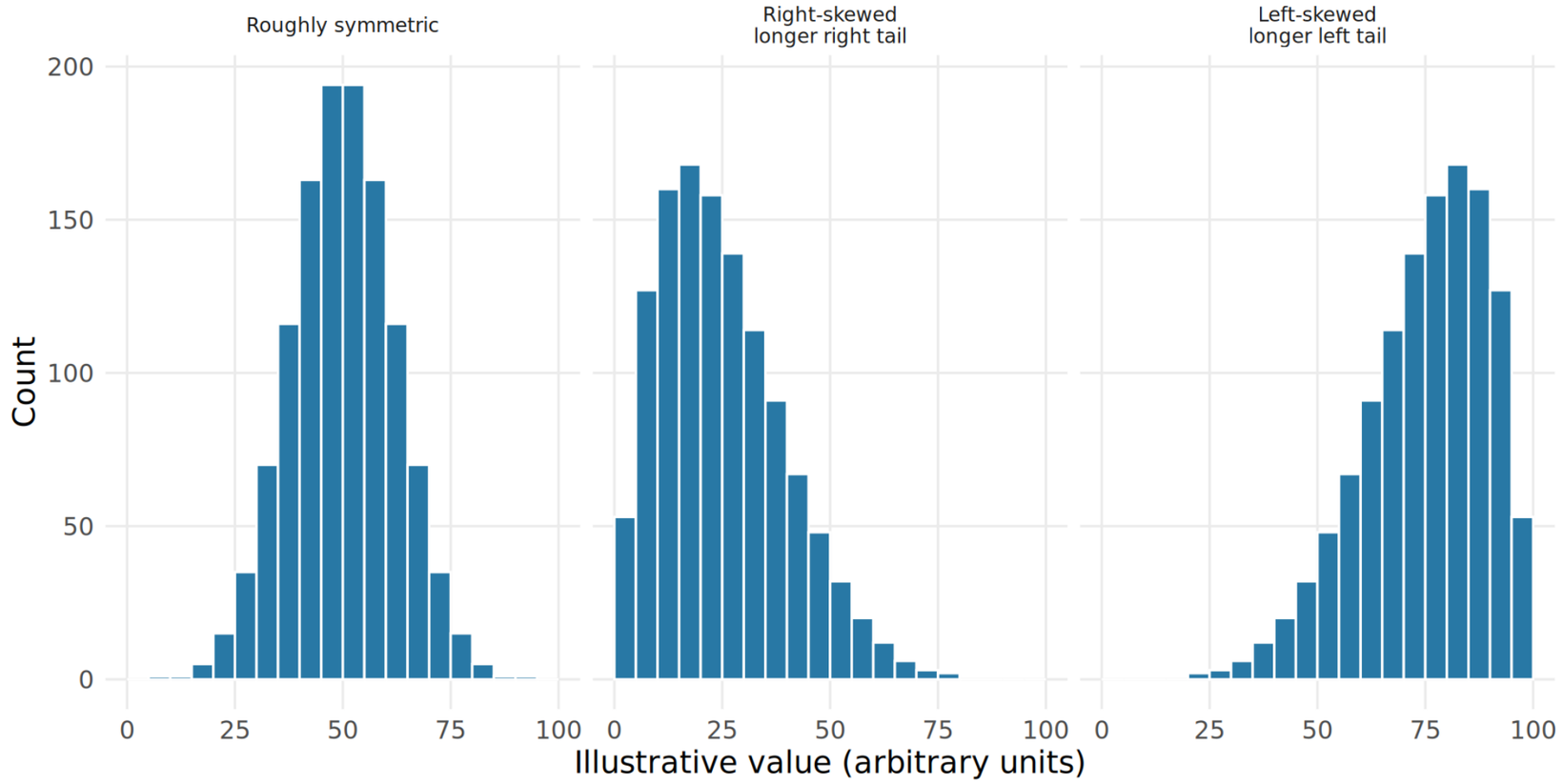
Four questions for a histogram

Shape: balance, tail direction, and prominent peaks?
Center: where is a typical region?
Spread: how far do values extend and concentrate?
Unusual features: gaps, isolated values, or small separated groups?

Read horizontal units and scale before comparing apparent widths.

Skewness follows the tail

Symmetric: similar sides. Right-skewed: longer right tail. Left-skewed: longer left tail.



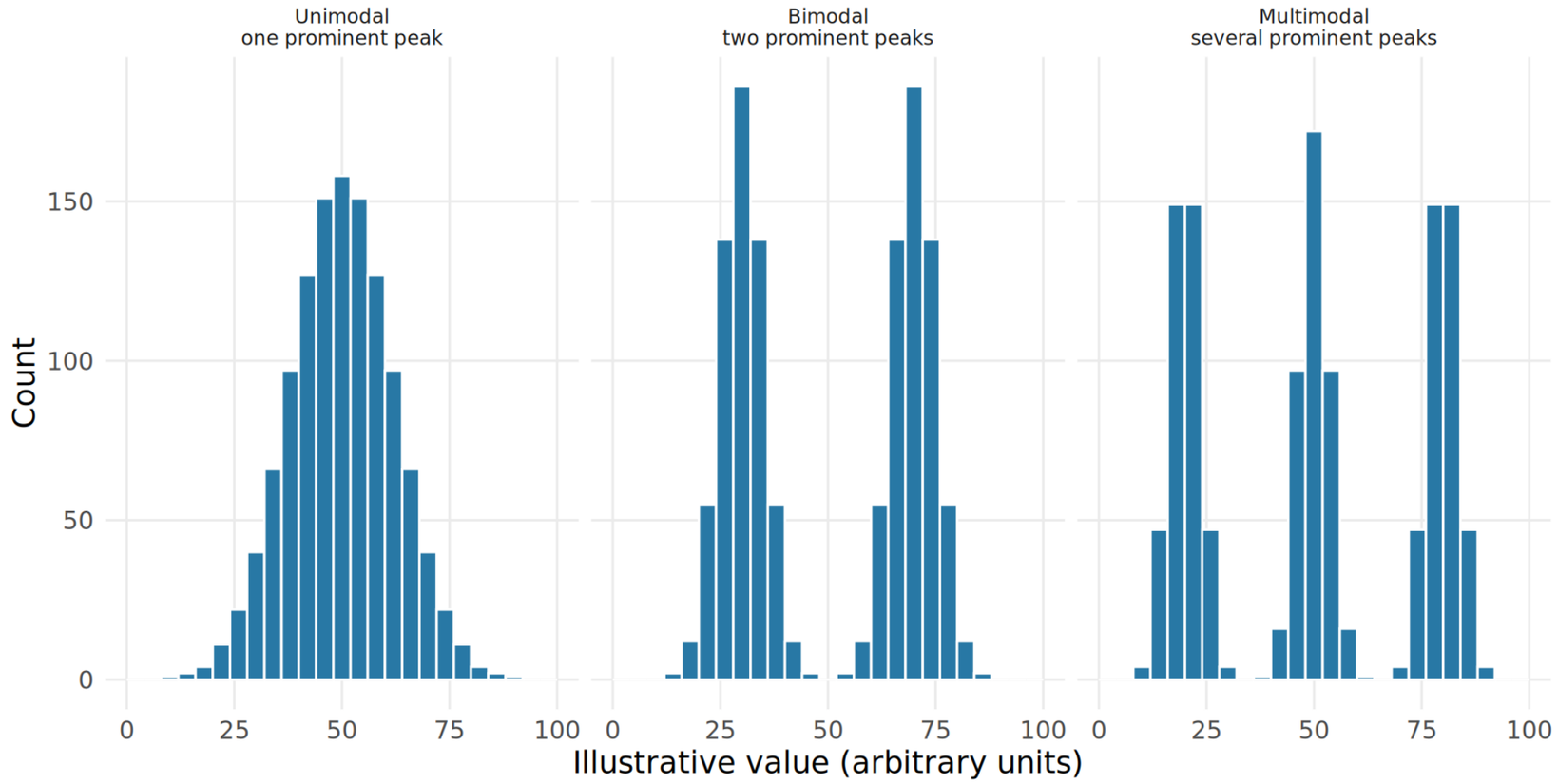
Constructed teaching illustrations, not observed Audit data or evidence of normality.

Shape needs careful language

Name skewness from the tail, not the tallest bar.
In many familiar one-peaked distributions, extremes pull the mean toward the tail.
That is not a universal mean/median ordering rule.
A symmetric histogram alone does not establish a normal distribution.

Prominent peaks and modality

Unimodal: one prominent peak. Bimodal: two. Multimodal: several.



Artificial teaching patterns. Small uneven bars are not automatically distinct modes.

Possible peaks need checking

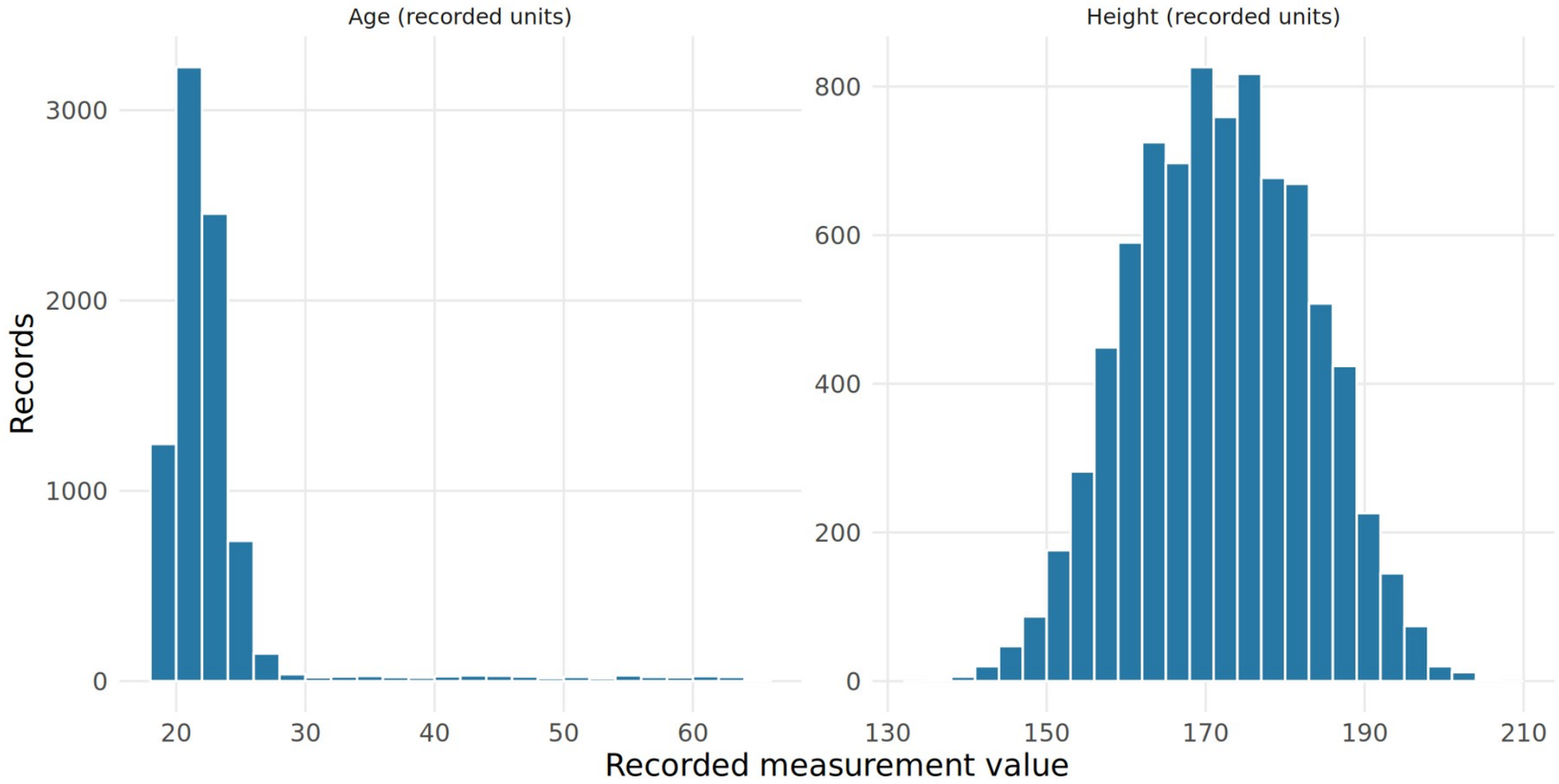
Several peaks can suggest mixed groups worth investigating.
They do not prove that those groups exist.
Sampling variation, rounding, bin width, and starting points affect the display.

The supplementary measurement file

The local file contains 8,239 age/height records.
It omits names and identifiers.
Population, collection method, real/simulated status, and units are undocumented.
Use recorded units and keep this separate from the textbook cases.

Your age-and-height comparison

Age has a long right tail; height has a more balanced central concentration.



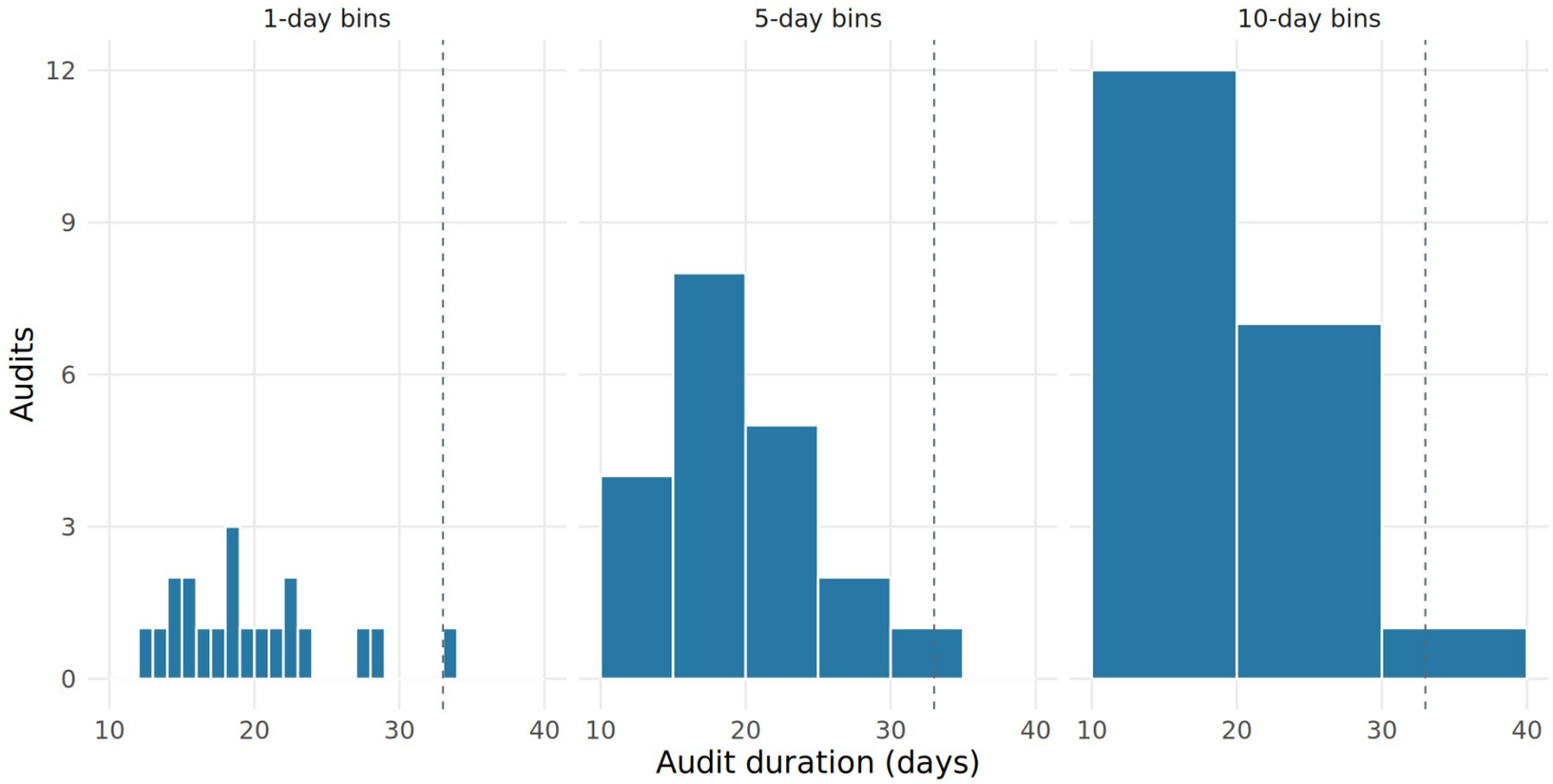
All 8,239 records remain. Different scales and bin widths prevent direct width/count comparisons.

Limits of the supplementary comparison

Small bumps do not establish subpopulations or exact normality.
Uncommon ages or heights are not automatically errors.
Do not treat this as a verified university survey or infer missing units.

Same audits, different bins

The same 20 values appear with 1-, 5-, and 10-day bins.



The dashed line marks 33 days, not an outlier threshold. Taller combined bins do not mean more audits overall.

A display choice can create a feature

Check several sensible widths and starting points.
Return to the raw values before naming a small peak or isolated group.
A histogram alone provides no universal outlier cutoff.
Lesson 3 introduces an explicit IQR screening rule.

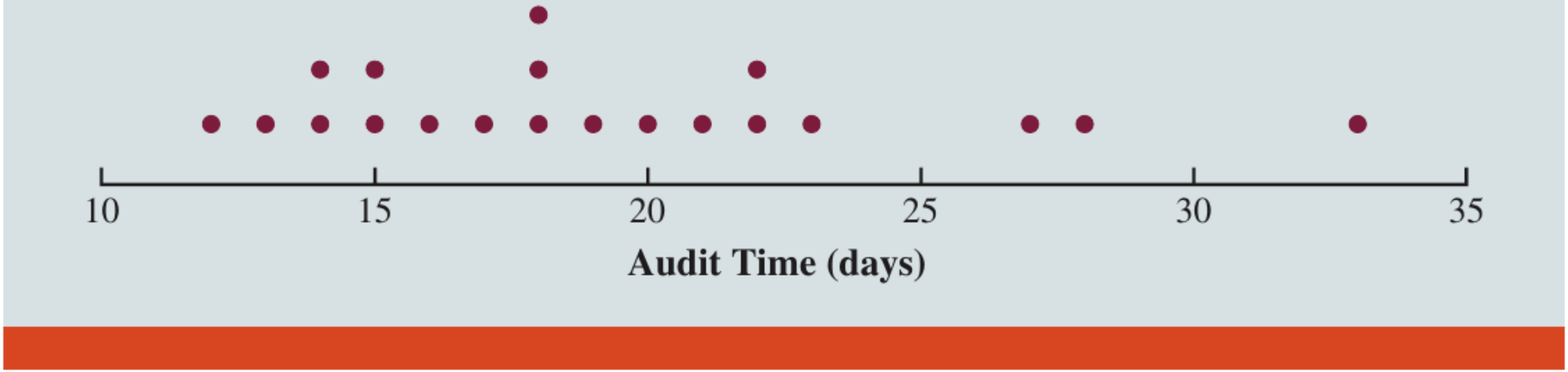
Audit interpretation in four sentences

Shape: a longer upper tail and one prominent peak in 5-day bins.
Center: high teens; exact mean 19.25 and median 18 days.
Spread: observed durations run from 12 to 33 days.
Unusual features: investigate the sparse upper end and 33-day audit.

The dot plot preserves individual values

One dot per audit; stacks preserve repeated durations.

FIGURE 2.11 Dot Plot for the Audit Time Data



Anderson et al., 8th ed., Figure 2.11, p.52. © 2020 Cengage Learning.

Bar chart or histogram?

Feature	Bar chart	Histogram
Horizontal meaning	Distinct categories	Numerical intervals
Order	Useful order; retain ordinal order	Numerical order
Separation	Gaps distinguish categories	Neighboring rectangles
Width	Consistent display choice	Represents interval width
Question	How common is each category?	How are values distributed?

Unequal class widths need density

With unequal widths, count heights can mislead.
Histogram area must represent frequency.
Density height = frequency / class width,
or relative frequency / class width.
Our equal-width examples can use count heights directly.

Cumulative frequencies

Add counts through successive ordered classes.
The denominator remains the same 20 audits.

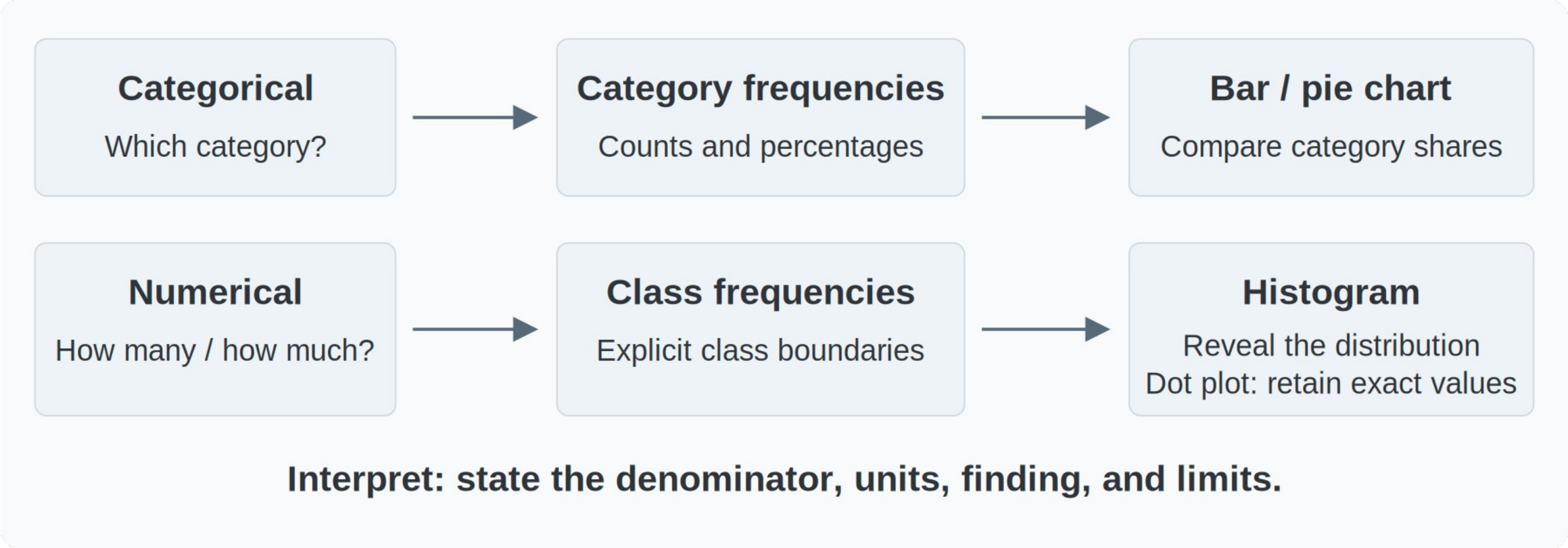
Duration threshold	Cumulative audits	Cumulative percent
Less than 15 days	4	20
Less than 20 days	12	60
Less than 25 days	17	85
Less than 30 days	19	95
Less than 35 days	20	100

17/20 = 85% took less than 25 days; 3/20 = 15% took at least 25 days.

A threshold inside a class

Grouped counts cannot identify exactly how many are below 18 days.
Return to the CSV for a threshold inside a class.
“Less than 20” and “20 or less” differ when a 20-day value is present.

Frequency workflow summary



Pause for the matching Quarto practice; keep meaning, construction, display, and interpretation aligned.

Data checks and limits

SoftDrink totals match after the documented name mapping; order is not asserted.
All 20 printed Audit observations match.
Shape/modality examples are artificial; student measurements have metadata limits.
The attributed dot plot remains distinct from original course R graphics.

Lesson 3: Numerical measures

Connect each measure to the distribution and the question.

Center, variability, and position

Interpret mean, median, and mode.
Explain what mean and range preserve and conceal.
Use variance, SD, quartiles, IQR, CV, and z-scores appropriately.
Compare distributions with consistent units and conventions.

A typical audit and its variability

Describe the same 20 year-end audit durations, in days.
What duration is typical, and how much do durations differ?

نصف المدد نفسها لعشرين عملية تدقيق لنهاية السنة، مقيسة بالأيام.
ما المدة المعتادة، وإلى أي حد تختلف مدد التدقيق؟

An observed sample describes these audits; it does not guarantee a future completion time.

Three meanings of typical

Mean balances all numerical values.
Median locates the middle after sorting.
Mode identifies the most frequent value or category.
Choose according to the question; the measures are not interchangeable.

The arithmetic mean

$\bar{x} = (\Sigma x_i) / n$

Add all observations, then divide by their number.

For Audit: total 385 days / 20 audits = 19.25 days.

Why the mean is useful

It compresses many observations into one summary of their level.
It keeps the original units.
Mean × number of observations = total.
Every value contributes, which matters for total workload.

What the mean leaves out

It does not show how values are distributed.
An extreme value can pull it away from most observations.
Sensitivity can matter for totals while weakening a typical-value description.
A mean needs a spread measure and a plot.

Median: the middle by rank

Sort the values before finding the middle.
With an even n, average the two middle numerical observations.
For 20 audits, positions 10 and 11 are both 18 days.
Median = (18 + 18) / 2 = 18 days.

With ties, “half below and half above” need not mean strictly below and strictly above.

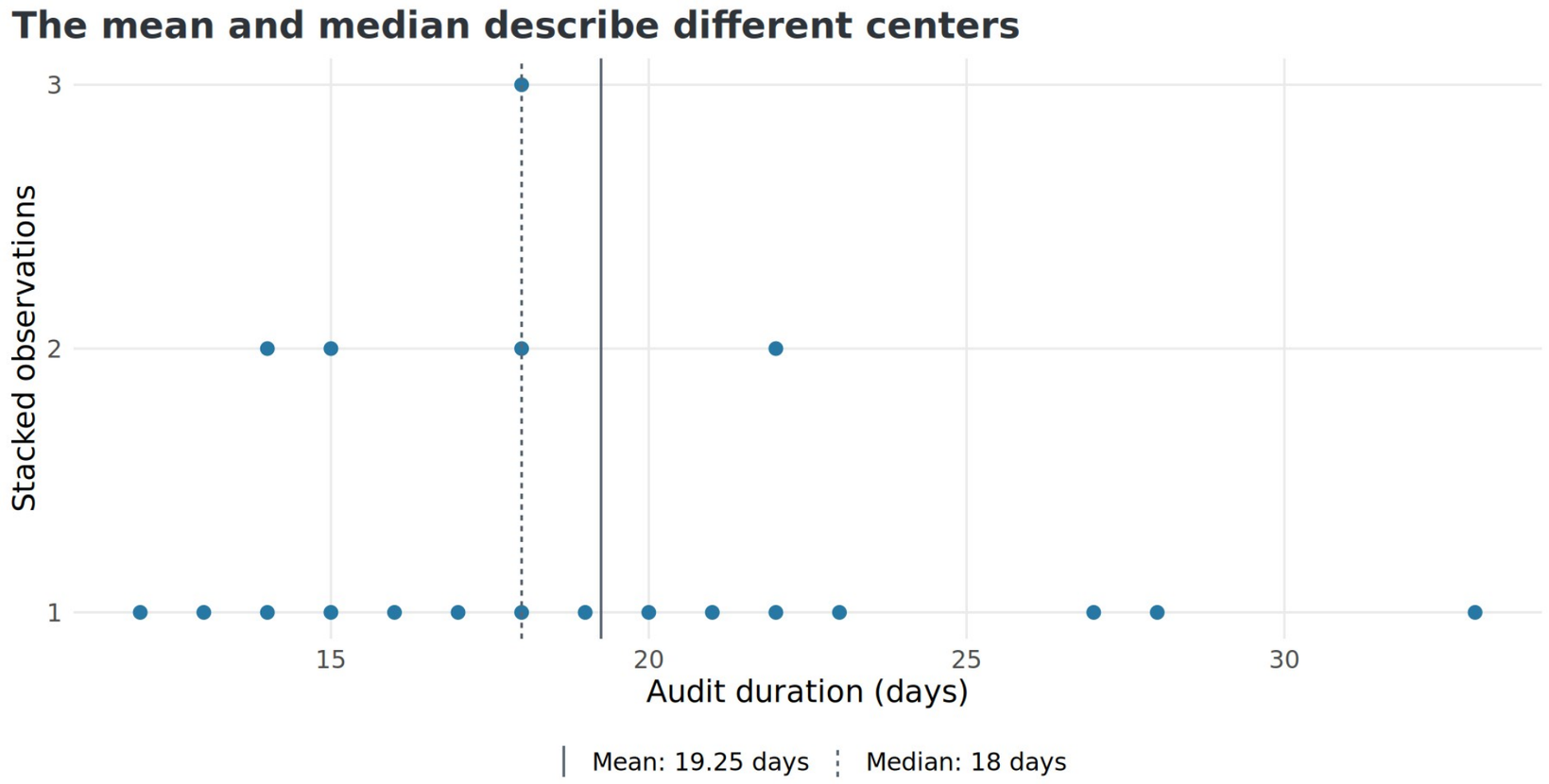
Mode: frequency rather than rank

The most frequent exact audit duration is 18 days.
A mode can be tied; report all equally most frequent values.
For categorical data, a modal category remains meaningful.
R’s mode() reports a storage mode, not the statistical mode.

Measure	Days
Mean	19.25
Median	18
Mode	18

Two centers in the Audit data

The mean is 19.25 days; the median is 18 days.



Longer durations pull the mean upward here. This ordering alone is not a universal skewness test.

Your seven-dollar example

Amounts: USD 1, 2, 3, 3, 3, 1, 100.
Total = USD 113. Mean = 113/7 ≈ USD 16.14.
The mean lies above six of the seven observations.
Median and mode are both USD 3.

Original instructor slide 38 / reference PDF p.51. No business setting is specified in the source.

Your original mean example

The Mean

- The **Arithmetic Mean** is the primary measure of central location.
- Generally, we refer to the arithmetic mean as simply the **Mean** or the **Average**.
- In order to calculate the mean of a data set, we simply add up the observations and divide by the number of observations in the population or sample.
- **However, the mean is sensitive to Outliers. Consider the following sample dataset:**

  Data = ($1.00, $2.00, $3.00, $3.00, $3.00, $1.00, $100.00)
- **Most of the data values are around $3, yet their average is $16.14**

Instructor-provided reference, original slide 38 (PDF p.51). The preceding slide gives the reduced explanation.

What this example does and does not show

The mean correctly relates the total to seven observations.
USD 3 better describes the middle or most frequent amount here.
Median and mode do not supply the total or describe spread.
Retain a valid USD 100 observation even when it makes the mean untypical.

Why center needs variability

A representative mean still tells us nothing by itself about separation.
Your five sets all have mean 200, despite different internal variation.

Set	Original values	Mean	Range
S1	100, 200, 300	200	200
S2	200, 200, 200	200	0
S3	0, 600, 0	200	600
S4	−600, 600, 600	200	1200
S5	−2000, 2500, 100	200	4500

Instructor slides 44–45 / PDF pp.60–61. Conceptual values in arbitrary units, not audit days.

The same mean can conceal different data

S2 has no variation: every observation is 200.
S5 runs from −2000 to 2500.
“Mean = 200” correctly summarizes a shared average.
It compresses away their differences. Describe center and dispersion together.

Range: the observed endpoints

Range = maximum − minimum.
For Audit: 33 − 12 = 21 days.
For S1–S5: 200, 0, 600, 1200, and 4500 units.
Useful for the full observed span, but it uses only two observations.

The USD 99 range

Return to USD 1, 2, 3, 3, 3, 1, 100.
Range = 100 − 1 = USD 99.
Six of seven observations occupy only USD 1 to USD 3.
Both the wide full span and tight main cluster are real features.

A hypothetical endpoint change

If USD 100 were USD 1000, the range would be USD 999.
The other six observations would remain exactly the same.
An endpoint change can dominate the range.

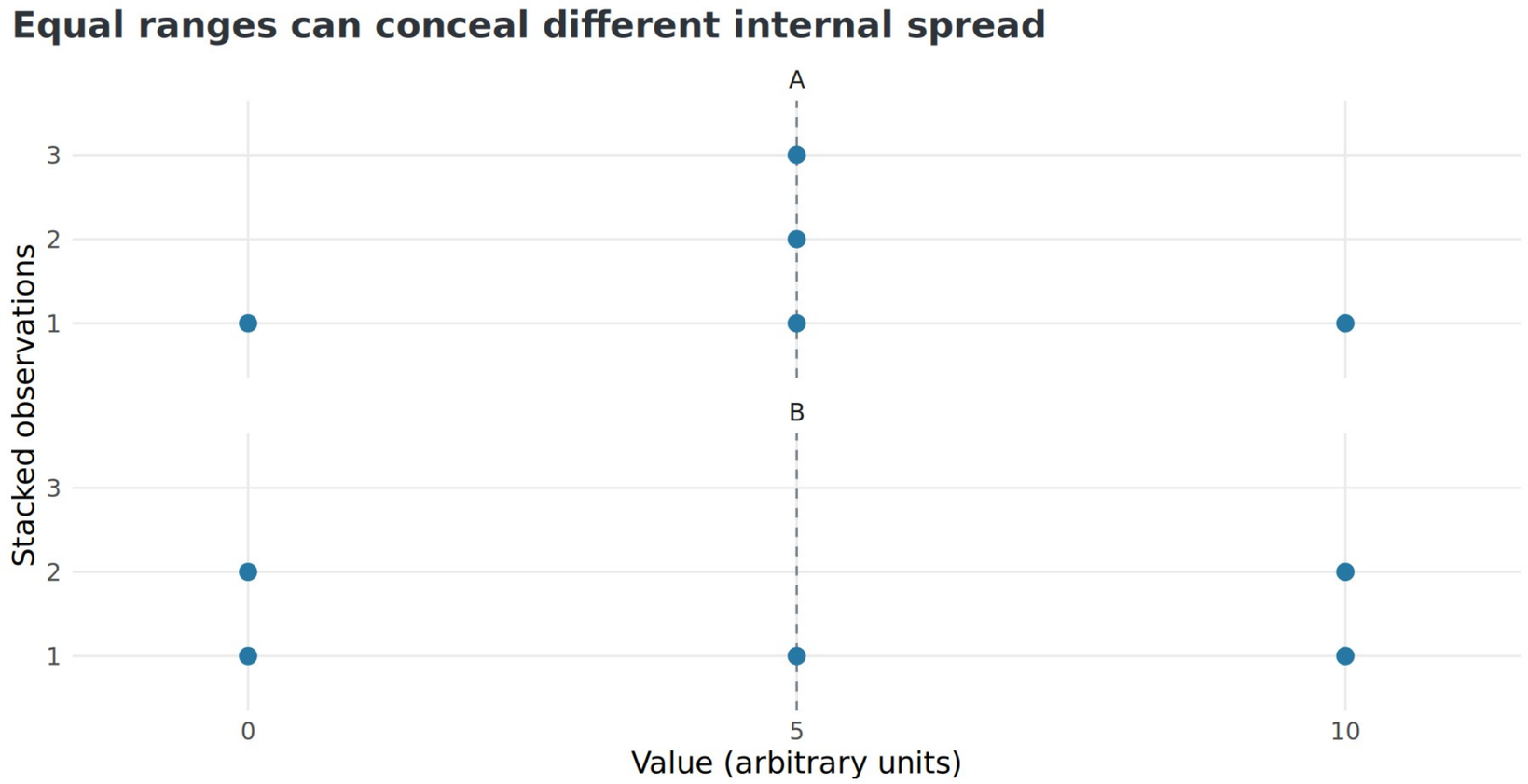
This is a labeled hypothetical sensitivity illustration. The CSV still contains USD 100.

Range misses internal differences

Original course illustration, arbitrary units:
A = 0, 5, 5, 5, 10.
B = 0, 0, 5, 10, 10.
Both have mean 5 and range 10. Their internal patterns differ.

Equal range, different internal spread

A concentrates at the center; B concentrates at the endpoints.



Original course illustration. Stacks preserve repeated observations; the dashed line is the shared mean.

Why use every deviation?

To describe internal spread, consider each distance from the mean.
Signed deviations sum to zero, so they cancel.
Absolute values prevent this cancellation.
Mean absolute deviation = $\Sigma|x_i - \bar{x}| / n$.

Mean absolute deviation for A and B

A’s distances: 5, 0, 0, 0, 5. Average = 2 units.

B’s distances: 5, 5, 0, 5, 5. Average = 4 units.

Every observation contributes, so the measure distinguishes internal spread.

```
# mean(v) is the reference center; subtraction gives every signed deviation.
# abs() takes magnitudes; the outer mean() averages those absolute distances.
mean(abs(v - mean(v)))
```

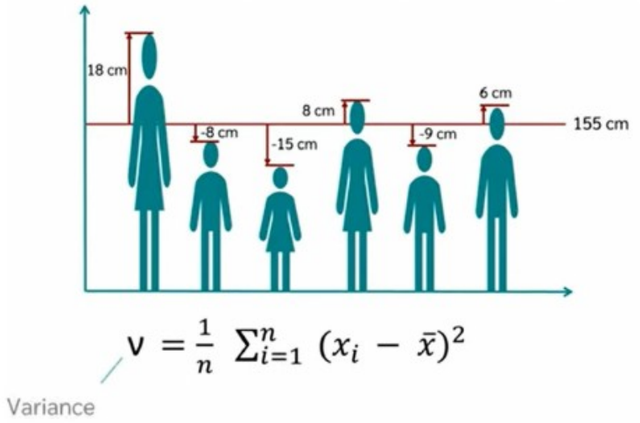
R’s mad() defaults to a scaled median absolute deviation, a different measure.

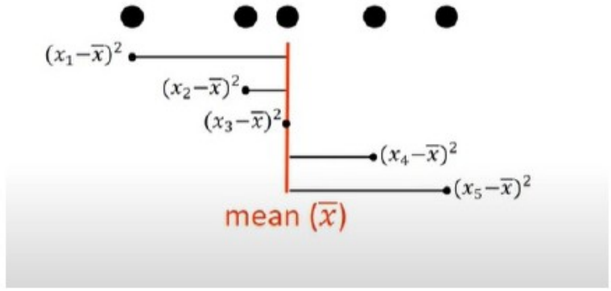
Absolute and squared distances

Absolute contributions grow linearly with distance.
Squared contributions emphasize larger distances more strongly.
This choice serves different purposes.
Neither approach is automatically resistant to extremes or universally better.

Your variance illustrations

Conceptual illustrations from your supplied variance/SD slide.





Original slide 47 / PDF p.63. The displayed 1/n formula is a population-style average; sample variance uses n − 1.

Sample variance

$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$, for $n > 1$.

Squaring prevents positive and negative deviations from cancelling.

Variance uses squared measurement units.

Estimating the mean uses one degree of freedom.

Sample or complete population?

A complete population variance uses its population mean and denominator N.
For these 20 audits, we use the sample denominator n − 1.
“All rows available” does not itself make the file a population.
The designation depends on the question.

Why the n − 1 adjustment?

Under independent sampling from a common population with finite variance,
the n − 1 adjustment makes sample variance an unbiased estimator.
This justification does not turn any convenience dataset into an independent sample.

Standard deviation

$s = \sqrt{s^2}$.

Audit sample variance ≈ 29.57 days². Sample SD ≈ 5.44 days.

SD summarizes spread around the mean in the original units.

Normality is not required merely to calculate it.

What SD does not mean

SD is not the standard error.
It is not the maximum possible deviation.
It does not guarantee that every observation lies within one SD of the mean.

Set	n	Mean	Range	Mean abs. deviation	Sample SD
A	5	5	10	2	3.54
B	5	5	10	4	5

A and B share n, mean, and range. Mean absolute deviation and SD distinguish their internal spread.

What improved, and what remains?

Using all values addresses the internal spread that range misses.
Mean absolute deviation and SD still respond to extreme values.
Next, quartiles and IQR describe the distribution’s central portion.

Percentiles and quartiles

Percentiles locate values in an ordered distribution.
Q1, Q2, Q3 correspond to the 25th, 50th, 75th percentiles.
Q2 is the median. Ties and interpolation matter in small samples.
State the convention used to compute sample percentiles.

The book’s percentile positions

Lp = (n + 1)p / 100, with interpolation between adjacent values.
For n = 20: L25 = 5.25, L50 = 10.5, L75 = 15.75.
The adjacent Audit values give Q1 = 15, Q2 = 18, Q3 = 22 days.

```
# quantile() locates the probabilities supplied by c().
# type = 6 follows the book’s interior-position rule; R defaults to type = 7.
quartiles <- quantile(x, c(.25, .50, .75), type = 6)
```

Consistent software conventions

R’s IQR() and ordinary ggplot2 boxplots have their own defaults.
Calculate IQR from the stated quartiles.
Use explicitly supplied boxplot statistics when matching type 6.
Agreement for Audit does not make methods interchangeable in every dataset.

Do not extrapolate the book’s interior-position rule beyond the available ordered values.

Interquartile range

IQR = Q3 − Q1 = 22 − 15 = 7 days.

It describes the width of the central portion.

With ties, an inclusive Q1–Q3 interval need not contain exactly half the records.

IQR is less sensitive to extremes, while leaving tail detail out.

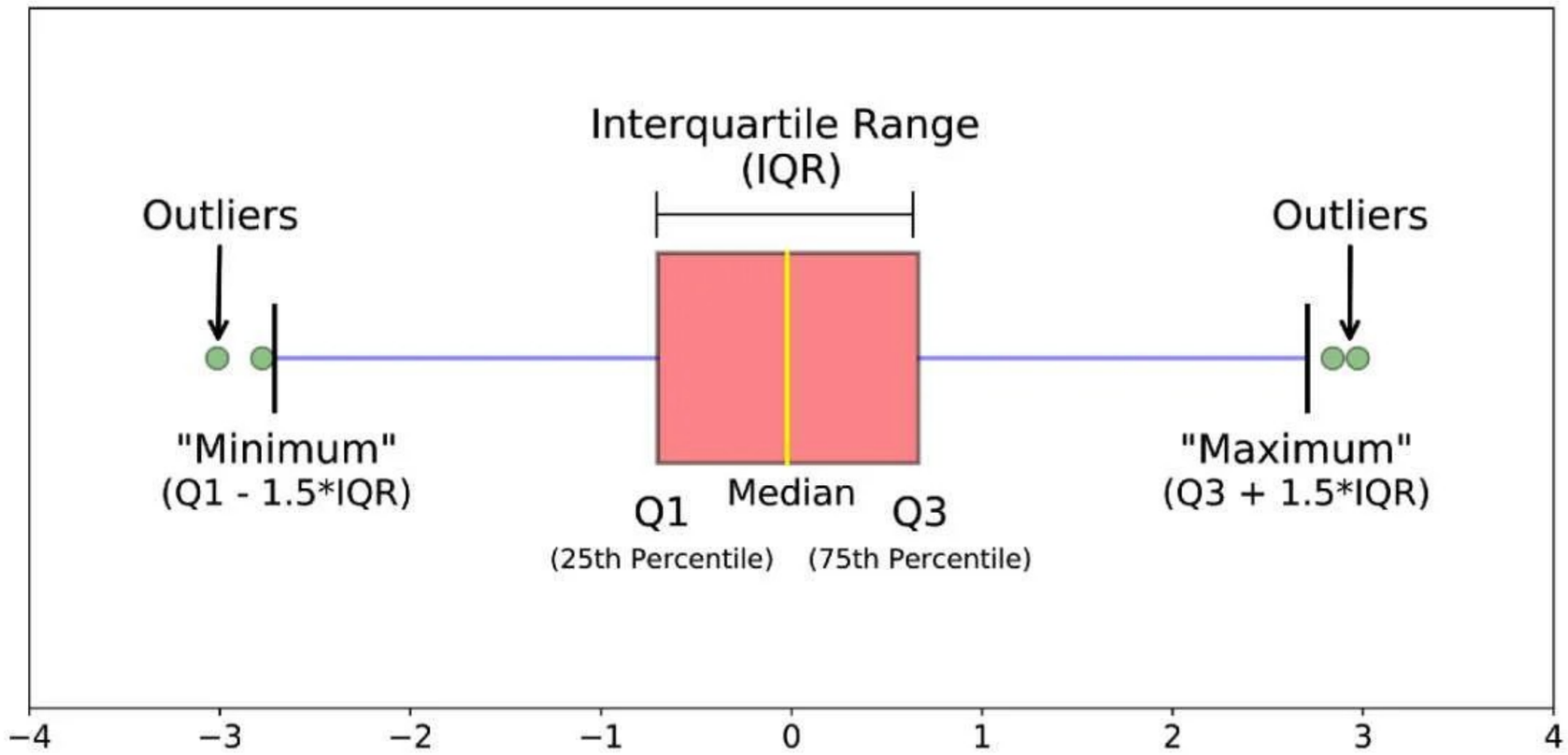
Five-number summary

Minimum, Q1, median, Q3, maximum.
For Audit: 12, 15, 18, 22, 33 days.

Boxplot screening rules

The box extends from Q1 to Q3 and marks the median.
Lower fence = Q1 − 1.5IQR. Upper fence = Q3 + 1.5IQR.
Whiskers stop at observed values inside the fences.
Beyond-fence observations remain part of the data.

Your boxplot anatomy diagram



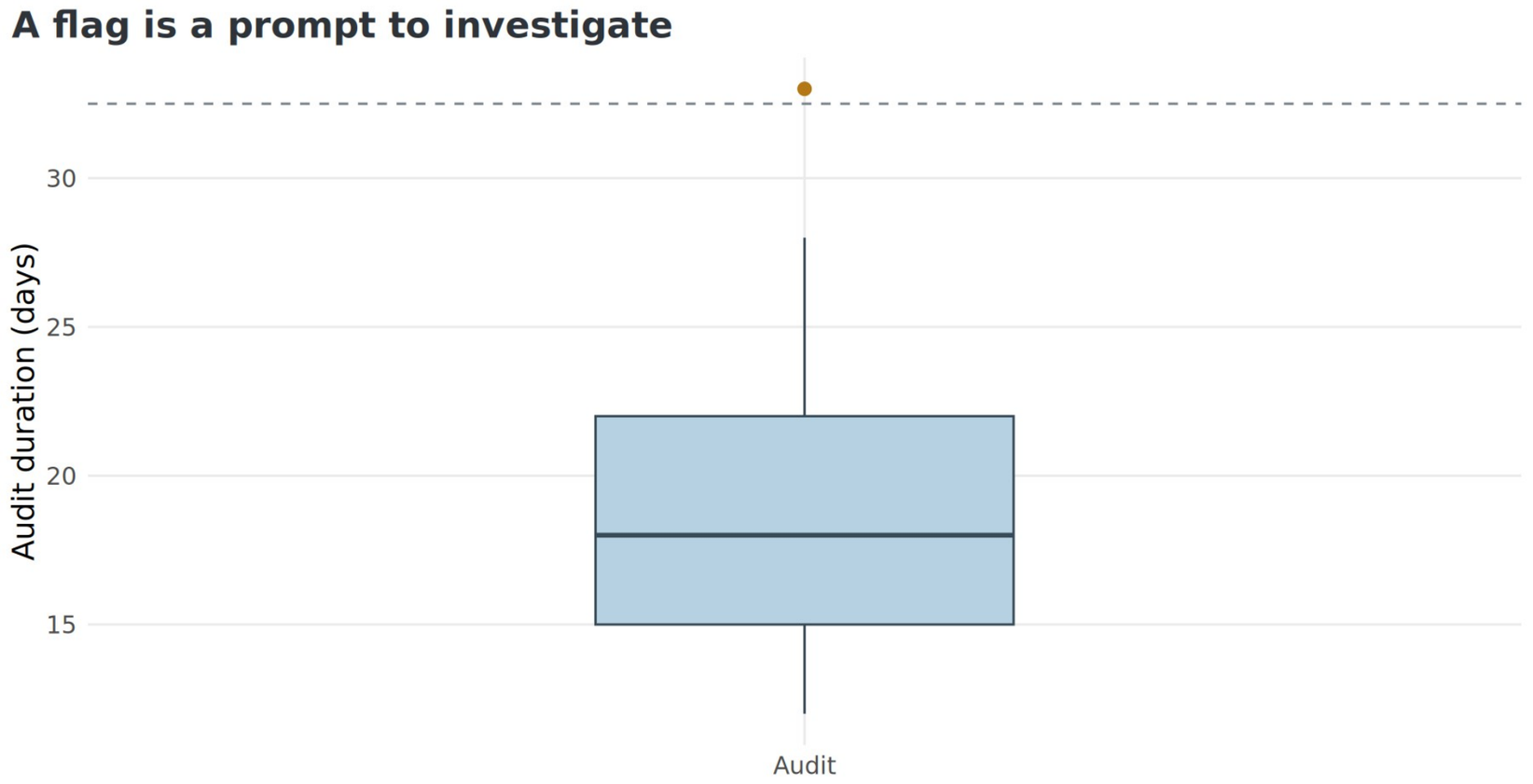
Supplied schematic. Its “Minimum” and “Maximum” formulas label fences, not generally observed endpoints.

Fences and whiskers are different

Audit fences: 4.5 and 32.5 days.
Observed whisker endpoints: 12 and 28 days.
The 33-day audit is flagged but remains in the dataset.
An outlier flag prompts investigation, not automatic deletion.

Audit boxplot with the stated convention

Explicit type-6 quartiles align the plot and numerical summary.



The five-number summary still includes the maximum 33, even though the upper whisker ends at 28.

A sensitivity comparison

Hypothetically replace the longest audit, 33 days, with 60 days.
Compare measures without changing the source data.

Dataset	Mean (days)	Median (days)	SD (days)	IQR (days)
Original observations	19.25	18	5.44	7
Hypothetical: 33 replaced by 60	20.60	18	10.25	7

Mean and SD change strongly; median and IQR remain unchanged in this illustration.

Relative spread and standardized position

CV describes spread relative to a suitable mean.
A z-score locates one value relative to the mean and SD.
They answer different questions and both need context.

Coefficient of variation

CV = 100s / $\bar{x}$, expressed as a percentage.
Compare relative spread for suitable ratio-scale variables.
Require a meaningful zero and a positive, nonzero mean.
A Celsius-temperature CV is not meaningful.

Audit CV ≈ 28.25%; a mean near zero makes the ratio unstable.

Z-scores

$z = (x - \bar{x}) / s$, with $s > 0$.

Sign indicates which side of the mean; magnitude gives distance in SD units.

For the 33-day audit, $z \approx 2.53$.

This calculation does not establish normality or an automatic outlier rule.

Compare Restaurant price distributions

Compare typical meal prices and variability across the 300 restaurants.
Ratings are ordered Good, Very Good, Excellent; prices are in USD.
One observation remains one restaurant.

نقارن أسعار الوجبات المعتادة ومدى تفاوتها في بيانات ثلاثمئة مطعم.
فئات الجودة مرتبة: جيد، وجيد جدًا، وممتاز؛ والأسعار بالدولار الأمريكي.
تظل كل مشاهدة ممثّلة لمطعم واحد.

These observational group comparisons describe association rather than a causal quality effect.

Group center and spread

Retain each group’s size alongside its summaries.

Quality	n	Mean (USD)	Median (USD)	SD (USD)
Good	84	19.73	19.50	6.13
Very Good	150	25.63	25	8.30
Excellent	66	34.35	34.50	8.29

Group quartiles and IQR

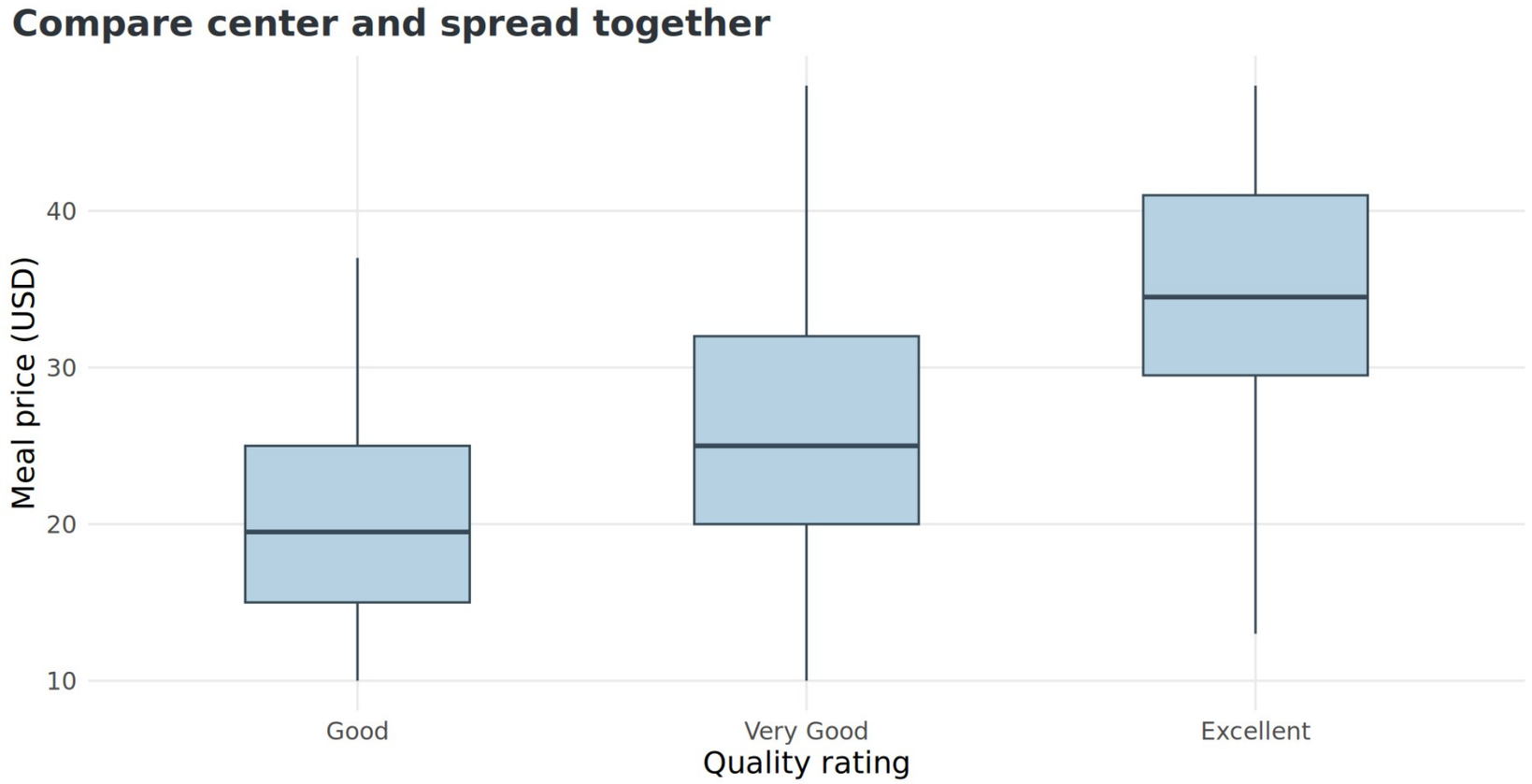
Use the same type-6 convention across every group.

Quality	Q1 (USD)	Q3 (USD)	IQR (USD)
Good	15	25	10
Very Good	20	32	12
Excellent	29.50	41	11.50

Good has IQR USD 10; Very Good has the largest IQR, USD 12.

Compare distributions together

Medians increase from USD 19.50 to 25.00 to 34.50, with overlap.



Higher typical price and greater variability are different questions. A category does not determine an individual price.

Numerical measures summary

Center	Variability	Position
What is typical?	How much do values differ?	Where does a value sit?
Mean · Median · Mode	Range · SD · IQR	Quartiles · Z-scores
Magnitude, rank, frequency	Read spread with center	Rank or SD units

Meaningful variables + units + conventions + a distribution plot

A summary describes the data; an unusual value needs investigation.

Describe center, variability, and position with units, conventions, and a plot.

A limitation suggests a companion

Mean pulled by extremes: add median and a distribution plot.
Center hides variation: add a spread measure.
Range misses internal values: add absolute deviation or SD.
SD responds to extremes: add IQR and the median.
Every companion still leaves some information out.

Pause for Lesson 3 practice. Preserve valid observations and explain the business meaning.

Your examples and their provenance

Seven dollar amounts: instructor slide 38 / reference PDF p.51.
Five equal-mean sets: slides 44–45 / PDF pp.60–61.
Absolute and squared deviation reasoning: slides 46–47.
The A/B equal-range comparison is an original course illustration.
The CSV and source notes preserve values, units, and origins.

Lesson 4: Relationships between variables

Choose tools for the variable pair.

State the comparison and its denominator.

From one variable to relationships

Compare category combinations, group distributions, and paired numbers.
Use a plot and numerical summaries together.
Describe association with its observational limits.

Match the pair to the question

Variable pair	Tools	Question
Categorical + categorical	Crosstab, percentages, bars	Which combinations occur?
Categorical + numerical	Group summaries, boxplots	How do distributions differ?
Numerical + numerical	Scatterplot, covariance, correlation	How do paired values move?

Restaurant: quality and price bands

Use 300 historical Los Angeles restaurant records.
Quality is ordered; meal price is measured in USD.
Ask how price bands differ within quality, and quality within price.

نستخدم ثلاثمئة سجل تاريخي لمطاعم في لوس أنجلوس.
فئات الجودة مرتبة، وسعر الوجبة مقيس بالدولار الأمريكي.
نسأل كيف تتوزع فئات السعر داخل كل مستوى جودة، وكيف تتوزع الجودة داخل كل فئة سعر.

Build a crosstab

Each cell counts one combination of quality and price band.
Bands are [10,20), [20,30), [30,40), [40,50) USD.
USD 20 belongs to the second band.

	10–<20	20–<30	30–<40	40–<50	Sum
Good	42	40	2	0	84
Very Good	34	64	46	6	150
Excellent	2	14	28	22	66
Sum	78	118	76	28	300

Row and column margins sum to 300. Each restaurant appears in one interior cell.

Ask “among whom?” first

64 restaurants are Very Good and priced USD 20–<30.
Joint: 64 / 300 = 21.33% of all restaurants.
Marginal: 150 / 300 = 50% are Very Good.
Within Very Good: 64 / 150 = 42.67%.
Within USD 20–<30: 64 / 118 = 54.24%.

Same numerator, different denominator: these percentages answer different questions.

Within a quality group

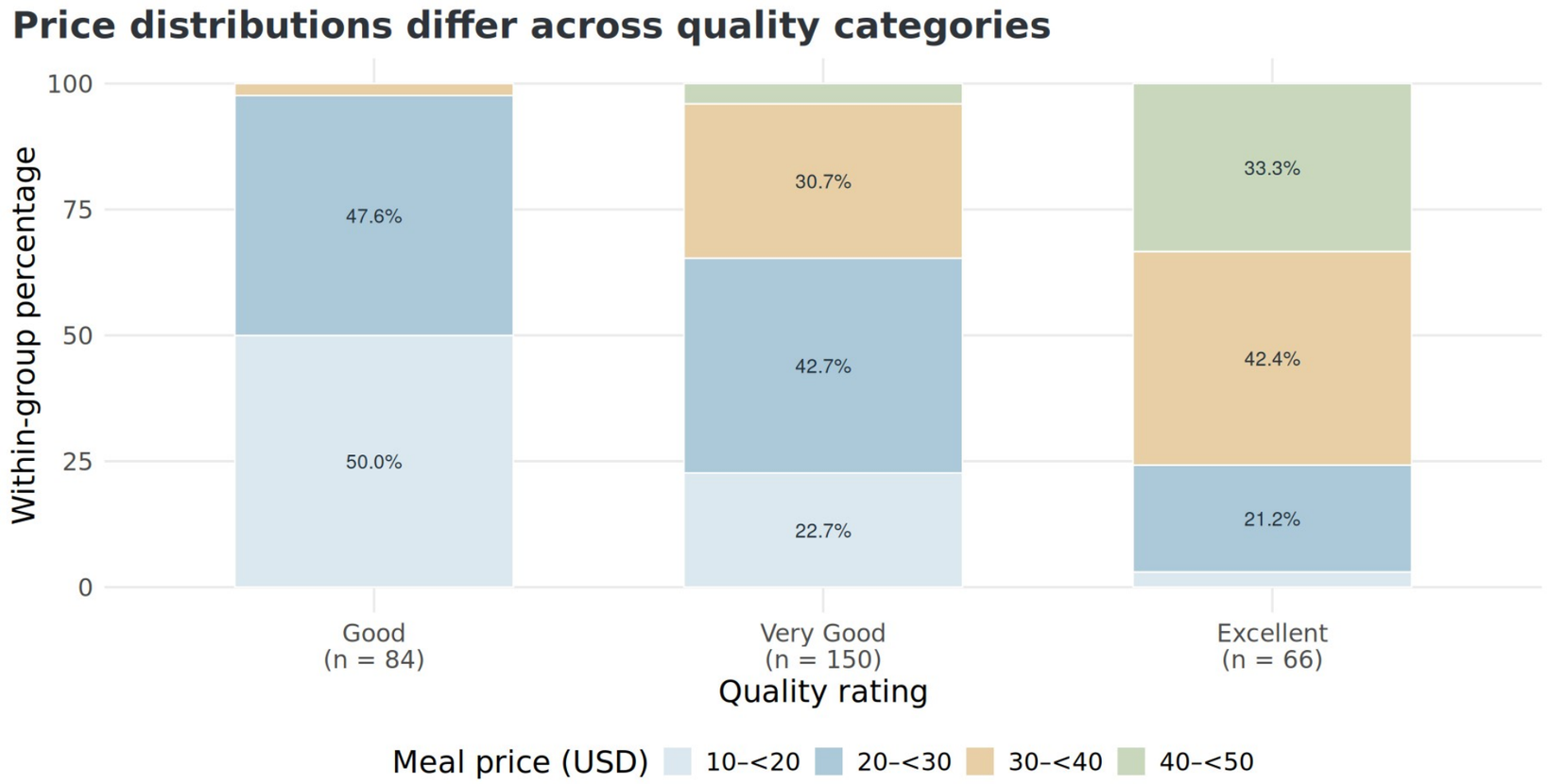
Divide each cell by its row total; each row sums to 100%.

	10–<20	20–<30	30–<40	40–<50	Total
Good	50	47.62	2.38	0	100
Very Good	22.67	42.67	30.67	4	100
Excellent	3.03	21.21	42.42	33.33	100

Half of Good restaurants are in USD 10–<20. Retain group sizes: 84, 150, and 66.

Compare price distributions within quality

Use a common 100% scale for groups of different sizes.



Compare precise values using the table; middle stacked segments have different baselines.

A threshold comparison

Among Excellent restaurants, 50 / 66 = 75.8% cost at least USD 30.
Among Good restaurants, 2 / 84 = 2.4% cost at least USD 30.
State the quality group in each denominator.
Calculate using counts, then round the displayed result.

Describe these records. Group differences alone do not show a causal quality effect.

Within a price band

Divide each cell by its column total; each column sums to 100%.

	10–<20	20–<30	30–<40	40–<50
Good	53.85	33.90	2.63	0
Very Good	43.59	54.24	60.53	21.43
Excellent	2.56	11.86	36.84	78.57
Total	100	100	100	100

Among USD 40–<50 restaurants, 22 / 28 = 78.6% are Excellent.

Reversing the condition changes the answer

Excellent among USD 40–<50: 22 / 28 = 78.6%.
USD 40–<50 among Excellent: 22 / 66 = 33.3%.
A crosstab reveals association, not causation.
Band choices and omitted subgroup structure can hide patterns.

These data do not establish that one variable causes the other.

Electronics: advertising and sales

Ten weekly pairs record television commercials and sales.
Sales are in hundreds of USD: 50 means USD 5,000.
Ask whether weeks with more commercials tend to have higher sales.

تسجّل عشرة أزواج أسبوعية عدد الإعلانات التلفزيونية والمبيعات.
المبيعات مقيسة بمئات الدولارات الأمريكية؛ فالقيمة 50 تعني 5000 دولار.
نسأل هل تميل الأسابيع ذات الإعلانات الأكثر إلى تسجيل مبيعات أعلى.

Checked against the local Stereo workbook. The proposed explanatory variable does not establish a cause.

The ten observed weekly pairs

One row pairs one week; sales are in hundreds of USD.

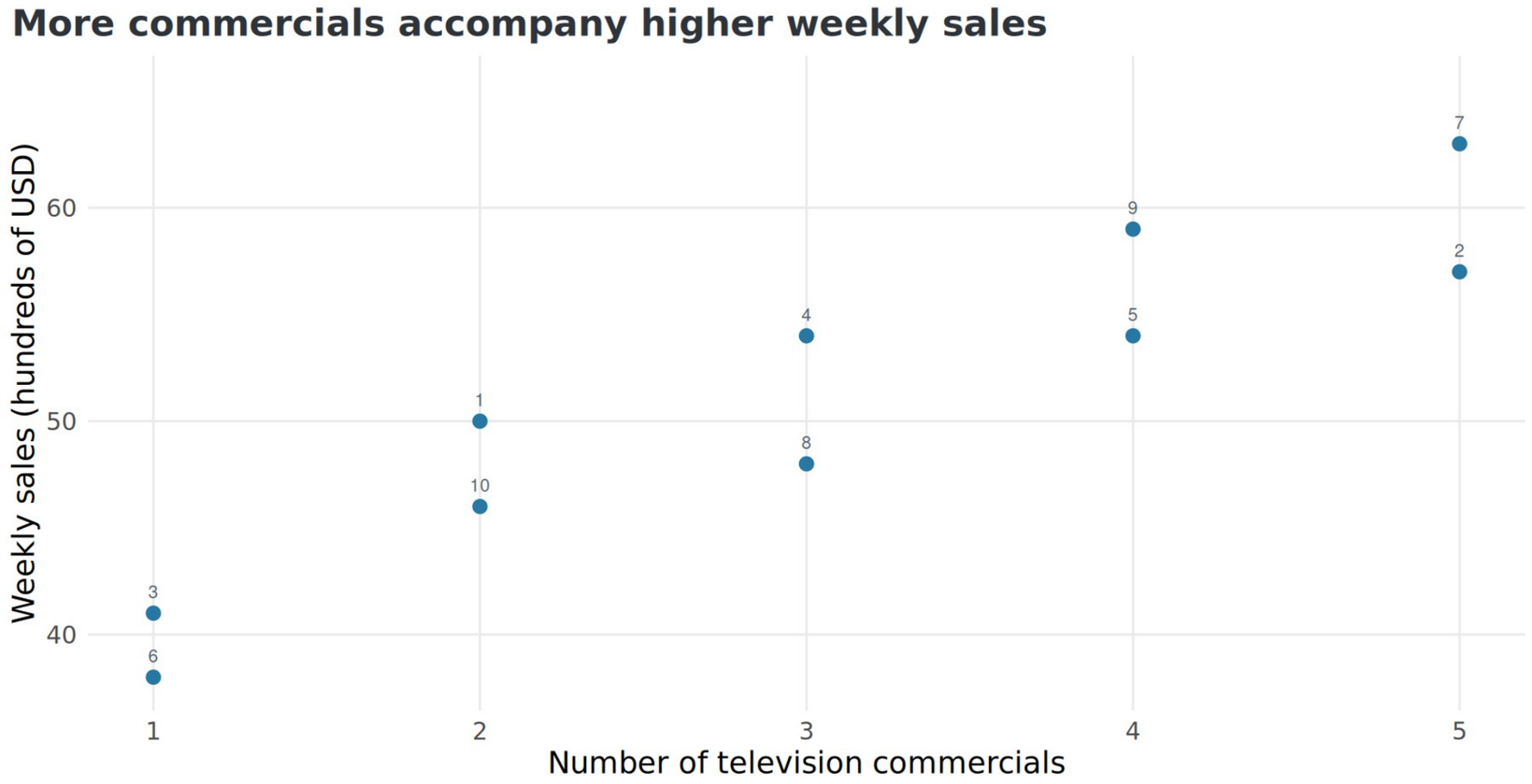
Week	Commercials	Sales (hundreds of USD)
1	2	50
2	5	57
3	1	41
4	3	54
5	4	54
6	1	38
7	5	63
8	3	48
9	4	59
10	2	46

Keep each week’s pair together

Place commercials on x and sales on y.
Sorting the columns independently destroys the pairs.
Do not connect points as a time path when describing this relationship.

Read the paired pattern

Positive direction; approximately linear form; fairly close clustering.



Describe variation and unusual observations as well as the overall direction.

Covariance starts with deviations

Compare each value with its own variable’s mean.
Same-sign deviations give a positive product.
Opposite-sign deviations give a negative product.
Sample covariance = $\Sigma[(x - \bar{x})(y - \bar{y})] / (n - 1)$.

Covariance for Electronics

$\bar{x}$ = 3 commercials; $\bar{y}$ = 51 hundreds of USD.
Sum of deviation products = 99.
Sample covariance = 99 / 9 = 11.
Units: commercials × hundreds of USD.
Its magnitude depends on units, so it is not a standardized strength score.

```
# cov() computes sample covariance using the paired numerical columns.
# Its denominator is n - 1; preserve week pairing and the documented sales units.
cov(electronics$commercials, electronics$sales_hundreds_usd)
```

Pearson correlation

r = sample covariance / ($s_x s_y$).

Both variables must have nonzero variation.

Unitless, from −1 to +1; sign gives linear direction.

Magnitude describes linear association, not every possible pattern.

Interpret the Electronics correlation

r ≈ 0.9305: a strong positive linear association in these ten weeks.
It is not a sales slope or “93% of sales.”
Read the scatterplot before describing the coefficient.
Calculation alone does not require normally distributed variables.

Inference and prediction need additional assumptions, addressed later in the course.

Change sales units

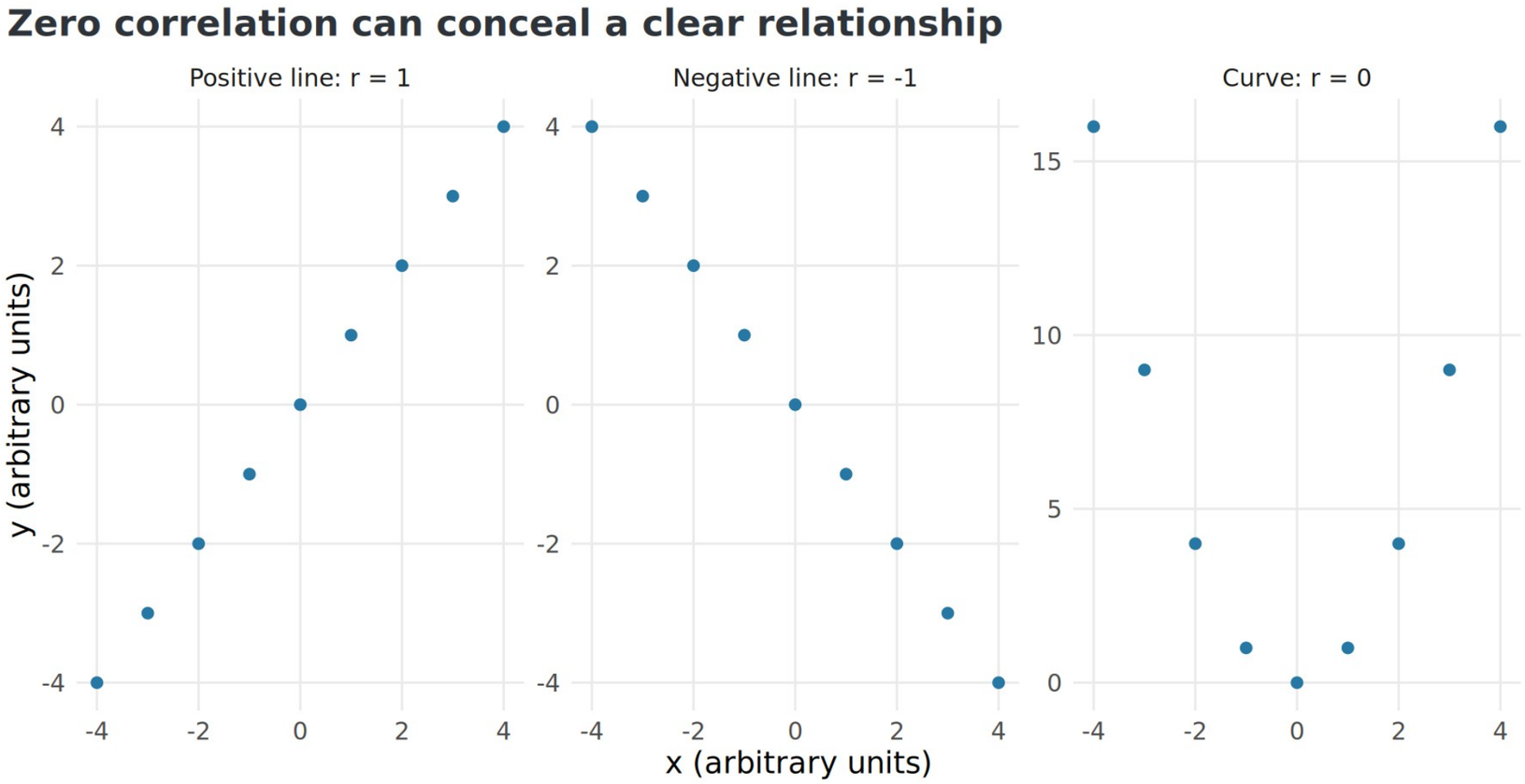
Convert hundreds of USD to USD by multiplying by 100.

Sales unit	Sample covariance	Correlation
Hundreds of USD	11	0.9305
USD	1100	0.9305

Positive unit rescaling changes covariance and SD proportionally; correlation stays the same.

The coefficient needs its plot

Artificial examples: increasing line, decreasing line, and a curved pattern.



The symmetric quadratic pattern has r = 0 despite a clear numerical relationship.

Limits of a single correlation

A near-zero r does not rule out a curved relationship.
Unusual points, restricted ranges, or mixed groups can change r.
A valid unusual observation is not automatically an error.
Other factors could affect advertising and sales.
Ten observed weeks do not guarantee the same future relationship.

A defensible relationship description

Name the variables, units, observational unit, and comparison.
Choose tools for the pair and report the denominator.
Describe form, direction, variation, and unusual observations.
Use summaries with the plot and state observational limits.

Pause for Lesson 4 practice: calculate, interpret, and qualify the result.

Relationship sources

Essentials of Modern Business Statistics with Microsoft Excel, 8th ed.
Restaurant and Electronics/Stereo use the checked teaching CSVs.
Figures and calculations retain the extended lesson’s data and conventions.
See its Sources and data section for the scope of each workbook match.

Lesson 5: Integrated review

Use the same sequence to produce a short reproducible report.

Your review report

Create one concise Quarto section per task.
Keep readable tables, plots, and business interpretations together.
State units and quartile conventions; preserve the source data.
Use the extended review for the complete submission instructions.

1. Define before summarizing

Identify the observational unit, variable meanings, and units.
Classify type and measurement level.
Distinguish counts from measured quantities.
Explain why averaging an ID or brand code is not meaningful.

2. SoftDrink frequencies

Build counts, relative frequencies, percentages, and a bar chart.
Identify the most frequent brand and interpret the denominator.
One recorded purchase does not represent every consumer’s preference.

3. Audit durations

Show the distribution and describe shape, center, spread, and unusual values.
Report n, mean, median, range, SD, and type-6 quartiles/IQR.
Apply fences as screening flags, retaining valid observations.
Choose a useful center/spread pair and explain its limitations.

4. Restaurant comparisons

Compare group n, medians, IQRs, and aligned boxplots.
Construct the four-band price crosstab.
Compare “at least USD 30” within quality using the correct denominators.
Separate an observed association from a causal claim.

5. Electronics association

Inspect the scatterplot, then calculate covariance and correlation.
Explain the units and the effect of changing sales units.
Describe linear direction and strength with the plot.
State what these ten weekly pairs cannot establish.

Before submitting

Meaning: are variables, units, and denominators clear?
Tools: do the displays and measures answer the question?
Reproducibility: are data paths and conventions stated?
Interpretation: does each number have a contextual explanation?
Limitations: what does the summary leave out?

The extended Quarto lessons remain the working material for code, practice, and full explanations.